Hardware Components Quick Reference (SGI™ 3000 Family)

SGI Confidential and Proprietary Information - For Internal Recipients Only

Document Number 108-0242-002

CONTRIBUTORS

Written by Jill Hietpas
Illustrated by Kelly Begley
Edited by Allison Gosbin
Production by Rhonda Kunsman
Engineering contributions by: Todd Andringa, Dick Brownell, Tom Hahmaks, Bill Henderson, Mitch Mueller, Steve Hein, Gary Spilde, Stan Demotts, Paul Pedersen, Mike Raske

INFORMATION CLASSIFICATION

LIMITED RIGHTS LEGEND

TRADEMARKS

Contents

1. **About This Booklet** 1-1
   - 1.1 Notational Conventions 1-1
   - 1.2 Supporting Documentation 1-1

2. **Configurations** 2-1
   - 2.1 Slot Numbering 2-2
   - 2.2 Rack Numbering 2-3
     - 2.2.1 Rack Placement 2-4
   - 2.3 Power Bay Configurations 2-5
     - 2.3.1 Brick Power Requirements 2-5
     - 2.3.2 Power Supply Configurations 2-6
   - 2.4 CPU and Memory Configurations 2-7
     - 2.4.1 Memory Configurations 2-7
   - 2.5 I- and P-brick Configuration and Slot Locations 2-11
     - 2.5.1 P- and I-brick VRM Locations 2-12
     - 2.5.2 Supported PCI Cards 2-13
     - 2.5.3 PCI Card Configuration Rules 2-15
     - 2.5.4 PCI Card Limits 2-15
     - 2.5.5 Adding PCI Cards in a System 2-15
       - 2.5.5.1 Using the PCI Slotting Matrix 2-19
   - 2.6 X-brick Configuration and Slot Locations 2-21
     - 2.6.1 X-brick Power Board and VRM Locations 2-21
     - 2.6.2 XIO Card Configuration Rule 2-22
     - 2.6.3 XIO Card Limits 2-22
   - 2.7 D-brick Configuration 2-22
     - 2.7.1 RAID Drive Configuration 2-23
     - 2.7.2 JBOD Drive Configuration 2-23

3. **Cabling and Connections** 3-1
   - 3.1 DNET Cable Connections 3-2
     - 3.1.1 NUMAlink 3 Cabling Guidelines 3-2
     - 3.1.2 Xtown2 Cabling Guidelines 3-2
   - 3.2 Console Connections and Global Master 3-3
   - 3.3 Brick Connectors 3-5
     - 3.3.1 C Brick 3-6
     - 3.3.2 R Brick 3-8
     - 3.3.3 P Brick 3-10
     - 3.3.4 X Brick 3-13
     - 3.3.5 I Brick 3-16
     - 3.3.6 D Brick (JBOD) 3-20
     - 3.3.7 D Brick (RAID) 3-21
     - 3.3.8 L2 Controller 3-23

3.3.9 Power Bay ... 3-24

4. **Commands** ... 4-1
4.1 L1 Commands ... 4-2
4.1.1 Virtual Debug Switch Settings ... 4-13
4.2 L2 Commands ... 4-15
4.3 PROM Monitor Commands ... 4-20
4.4 POD Commands ... 4-23
4.5 IRIX Commands ... 4-28

5. **System Procedures** ... 5-1
5.1 Establishing System Controller Connections ... 5-2
5.1.1 SGI 3200 Systems ... 5-2
5.1.1.1 Serial Connection to the L1 Controller ... 5-2
5.1.1.2 USB Connection to the L2 Emulator ... 5-2
5.1.2 SGI 3400 and 3800 Systems ... 5-3
5.1.2.1 Serial Connection to the L2 Controller ... 5-3
5.1.2.2 Ethernet Connection to the L2 Controller ... 5-3
5.2 Flashing System Controller Firmware ... 5-5
5.2.1 Obtain Updated L3 Software Revisions ... 5-5
5.2.2 Flash L1 Firmware ... 5-6
5.2.3 Flash L2 Firmware ... 5-6
5.3 Flashing IP35 Firmware ... 5-7
5.3.1 Flashing IP35 Firmware from a CD-ROM ... 5-7
5.3.2 Flashing IP35 Firmware from the PROM Monitor ... 5-7

6. **Troubleshooting** ... 6-1
6.1 Interpreting the LEDs Command Values ... 6-2
6.2 Cable Pinouts ... 6-8
6.2.1 48-V Power Connector ... 6-8
6.2.2 DNET Cable Connector and Pinouts ... 6-10
6.3 NASID to CPU Decoding ... 6-12
6.3.1 3200 Systems ... 6-12
6.3.2 3400 Systems ... 6-12
6.3.3 3800 Systems ... 6-13
6.4 Troubleshooting a System Hang ... 6-13

7. **FRU and Parts List** ... 7-1
7.1 C Brick (013-3128-003) ... 7-2
7.2 I Brick (013-3035-002) ... 7-3
7.3 R Brick (013-3037-002) ... 7-5
7.4 P Brick (013-3034-002) ... 7-6
7.5 X Brick (013-3036-003) ... 7-8
7.6 D Brick ... 7-9
7.7 G Brick ... 7-10
7.8 Cables ... 7-11

7.9 Power Components 7-11
7.10 Misc. Support Equipment 7-12
7.11 Brick Hardware 7-12
7.12 Rack and Trim, 39U 7-12
7.13 Rack and Trim, 17U 7-14
7.14 PCI Options 7-15
7.15 XIO Options 7-15
7.16 Tools List 7-16

7. Notes 8-1

Figures

Figure 2-1	Slot Numbering	2-2
Figure 2-2	Rack Numbering	2-3
Figure 2-3	Distributed Power Supply Configuration (front)	2-6
Figure 2-4	Address Bit Layout	2-8
Figure 2-5	C-brick DIMM, PIMM, and VRM Locations	2-10
Figure 2-6	P- and I-brick Slot Locations	2-11
Figure 2-7	I-brick Power Board and VRM Locations	2-12
Figure 2-8	P-brick Power Board and VRM Locations	2-12
Figure 2-9	PCI Slotting Mode	2-16
Figure 2-10	Adding PCI Cards to a System in Loose-pack Mode	2-17
Figure 2-11	Adding PCI Cards to a System in Close-pack Mode	2-18
Figure 2-12	X-brick Physical Slot Numbering	2-21
Figure 2-13	X-brick Power Board and VRM Locations	2-21
Figure 2-14	Disk Enclosure Numbering Sequence	2-22
Figure 2-15	D-brick (RAID) Channel Assignments (fully populated D brick)	2-23
Figure 2-16	D-brick (RAID) Channel Assignments	2-23
Figure 2-17	JBOD Drive Configuration	2-24
Figure 3-1	DNET Cable Labels	3-2
Figure 3-2	3200 Console Connection and Global Master	3-3
Figure 3-3	3400 and 3800 Console Connections and Global Master	3-4
Figure 3-4	`hinv` and `ioconfig.cfg` Screen Format for an I Brick	3-5
Figure 3-5	C-brick Connectors	3-6
Figure 3-6	R-brick Connectors	3-8
Figure 3-7	P-brick Connectors	3-10
Figure 3-8	X-brick Connectors	3-13
Figure 3-9	I-brick Connectors	3-16
Figure 3-10	D-brick Connectors	3-20
Figure 3-11	D-brick Connectors (RAID)	3-21
Figure 3-12	L2 Controller Connectors	3-23
Figure 3-13	Power Bay Connectors	3-24
Figure 4-1	L1 Prompt	4-2
Figure 4-2	POD Prompt	4-23
Figure 6-1	48-V Power Connector Pinouts (from Rear of Power Bay)	6-8
Figure 6-2	DNET Receptacle Pinouts (R Brick)	6-10
Figure 6-3	DNET Cable Connector Pinouts	6-11
Figure 6-4	Flowchart for Troubleshooting a System Hang	6-14
Figure 6-5	Troubleshooting Flowchart Continued (refer to Figure 6-4)	6-15

x

Tables

Table 1-1	Supporting Documentation	1-2
Table 2-1	Compute Rack Placement	2-4
Table 2-2	Brick Power Requirements	2-5
Table 2-3	Power Bay Configuration	2-6
Table 2-4	PDU Power-phase Configuration	2-7
Table 2-5	DIMM Configurations	2-7
Table 2-6	CPU Slice and Console Subchannel Numbers	2-7
Table 2-7	Memory Error Bit Definitions	2-8
Table 2-8	Memory Remapping	2-9
Table 2-9	Supported PCI Cards	2-13
Table 2-10	PCI Card Weighted Values	2-15
Table 2-11	PCI Slotting Matrix	2-20
Table 2-12	XIO Card System Limits	2-22
Table 3-1	3400 and 3800 NASID Numbering	3-3
Table 3-2	C-brick Connectors	3-6
Table 3-3	R-brick Connectors	3-9
Table 3-4	Screen Output Showing P-brick Configuration	3-11
Table 3-5	P-brick Connectors	3-11
Table 3-6	Screen Output Showing X-brick Configuration	3-13
Table 3-7	X-brick Connectors	3-14
Table 3-8	Screen Output Showing I-brick Configuration	3-17
Table 3-9	I-brick Connectors	3-17
Table 3-10	D-brick Connectors (JBOD)	3-20
Table 3-11	D-brick Connectors (RAID)	3-22
Table 3-12	L2 Controller Connectors	3-23
Table 3-13	Power Bay Connectors	3-24
Table 3-14	Power Supply LED States	3-25
Table 4-1	Connecting to the L1	4-2
Table 4-2	L1 Commands (Version 1.4.1)	4-2
Table 4-3	Virtual Debug Switch Settings	4-13
Table 4-4	Connecting to the L2	4-15
Table 4-5	L2 Commands (Version 1.4.2)	4-15
Table 4-6	PROM Monitor Commands (Version 6.54)	4-20
Table 4-7	POD Mode Command Reference (Version 6.54)	4-23
Table 4-8	Useful IRIX Commands	4-28
Table 5-1	*flashsc* and *l2term* Command Options	5-5
Table 6-1	Useful POD Codes	6-2
Table 6-2	LED Command Values and Descriptions	6-2
Table 6-3	DC Power Port Connector Pinouts	6-8
Table 6-4	3200 System NASID to CPU Decoding	6-12
Table 6-5	3400 System NASID to CPU Decoding	6-12
Table 6-6	3800 System NASID to CPU Decoding	6-13
Table 7-1	C-brick FRUs	7-2

Table 7-2	I-brick FRUs	7-3
Table 7-3	R-brick FRUs	7-5
Table 7-4	P-brick FRUs	7-6
Table 7-5	X-brick FRUs	7-8
Table 7-6	D-brick FRUs	7-9
Table 7-7	G-brick FRUs	7-10
Table 7-8	Cable FRUs	7-11
Table 7-9	Power Component FRUs	7-11
Table 7-10	Support Equipment	7-12
Table 7-11	Brick Hardware	7-12
Table 7-12	Rack and Trim Components (39 U)	7-12
Table 7-13	Rack and Trim Components (17 U)	7-14
Table 7-14	PCI Options	7-15
Table 7-15	XIO Options	7-15
Table 7-16	Tools List	7-16

Record of Revision

Version	Description
001	February 2001 Original printing
002	April 2001 *(Note: the changes in this review were not technically reviewed.)* • Updated L1 commands to support firmware release 1.4.1 • Updated L2 commands to support firmware release 1.4.1 • Removed 3800 rack placement table and DIMM color table. • Added PCI card matrix (Table 2-9). • Added flowchart for Troubleshooting a System Hang to support PRB 200235

Chapter 1

About This Booklet

This booklet contains quick-reference information for experienced SGI field service personnel who maintain an SGI Origin 3000 series server. This chapter provides the notational conventions and a list of supporting documentation for you to refer to if you need more detail or additional information.

1.1 Notational Conventions

Convention	Meaning
`screen display`	Fixed-space font denotes literal items such as references to commands, files, routines, path names, output, messages, and programming language structures
variable	Italic typeface denotes variable entries and words or concepts that are being defined such as command variables and references to document titles.
`user input`	Bold, fixed-space font denotes literal items that you enter on the keyboard in interactive sessions.
...	Ellipses indicate that one or more elements have been removed and/or that the preceding element can be repeated
[]	Square brackets enclose optional portions of a command
<>	Pointed brackets enclose required portions of a command
|	Dividers are used to separate more than one possible value for commands and variables

1.2 Supporting Documentation

Table 1-1 lists the reference documentation that supports the information in this booklet. If you need more detail than what is provided here, refer to these documents.

Table 1-1 Supporting Documentation

Chapter	Supporting Document	URL
Configurations	SGI Origin 3000 Series Technical Configuration Manuals	http://wwwcf.americas.sgi.com/PUBLIC/tech_pub/
Cables and Connections	System Cabling	http://servinfo.csd.sgi.com/sgi3000/sysmaint.htm
	System Components	http://servinfo.csd.sgi.com/sgi3000/prodover.htm
Commands	System Controllers	http://servinfo.csd.sgi.com/sgi3000/prodover.htm
	IP35 and IO7 Commands Reference	http://servinfo.csd.sgi.com/sgi3000/sysmain.htm
	Power-on Diagnostics and Discovery	http://servinfo.csd.sgi.com/sgi3000/sysmaint.htm
Procedures	Hardware Replacement Procedures	http://servinfo.csd.sgi.com/sgi3000/sysmaint.htm
	Service Procedures	TBD
FRU List	FRU and Tools Plan	http://www-logistics.csd/cf/prod/spares/o3kspare.htm
	Illustrated Parts Catalog	http://servinfo.csd.sgi.com/sgi3000/sysmaint.htm

Chapter 2

Configurations

Subsection	Page Number
Slot Numbering	page 2-2
Rack Numbering	page 2-3
Rack Placement	page 2-4
Power Bay Configurations	page 2-5
Brick Power Requirements	page 2-5
Power Supply Configurations	page 2-6
CPU and Memory Configurations	page 2-7
I- and P-brick Configuration and Slot Locations	page 2-11
P- and I-brick VRM Locations	page 2-12
PCI Card Configuration Rules	page 2-15
PCI Card Limits	page 2-15
Adding PCI Cards in a System	page 2-16
Using the PCI Slotting Matrix	page 2-19
X-brick Configuration and Slot Locations	page 2-21
X-brick Power Board and VRM Locations	page 2-21
XIO Card Configuration Rule	page 2-22
XIO Card Limits	page 2-22
D-brick Configuration	page 2-22

2.1 Slot Numbering

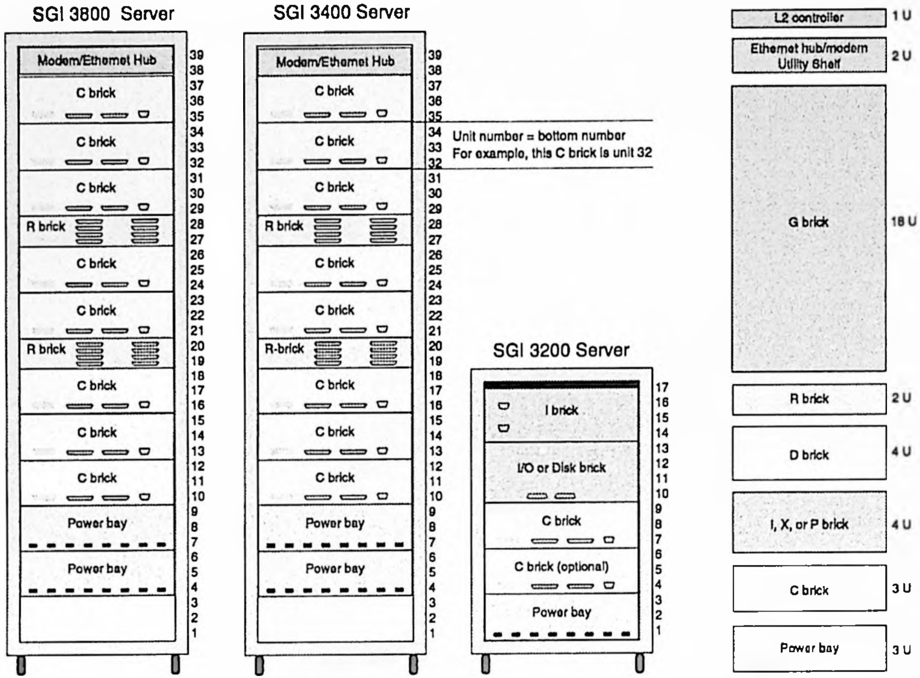


Figure 2-1 Slot Numbering

2.2 Rack Numbering

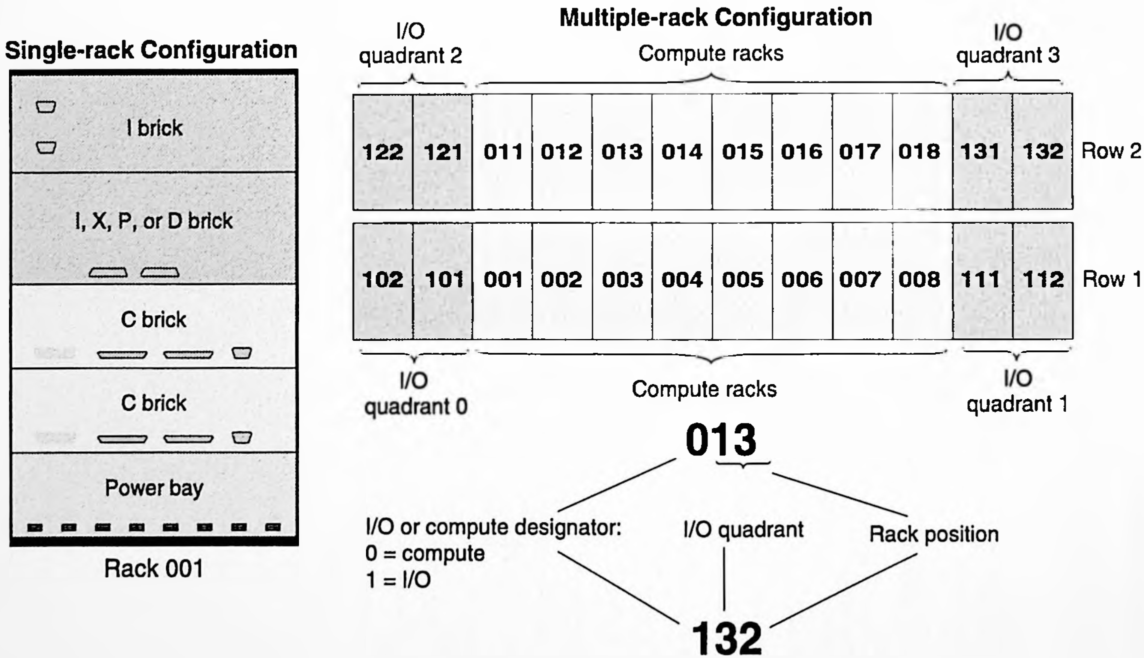


Figure 2-2 Rack Numbering

2.2.1 Rack Placement

Refer to Table 2-1 for the order in which to add compute racks to a system.

Note: If you are upgrading to a 160-processor configuration from a 128-processor configuration, move rack 004 to position 012 and add rack 013.

Table 2-1 Compute Rack Placement

Number of Racks	Number of Processors (max)	Rack number (Row 1)	Rack Number (Row 2)
1	32	001	
2	64	001—002	
3	96	001—003	
4	128	001—004	
5	160	001—003	012—013
6	192	001—004	012—013
7	224	001—004	011—013
8	256	001—004	011—014
9	288	001—005	011—014
10	320	001—005	011—015
11	352	001—006	011—015
12	384	001—006	011—016
13	416	001—007	011—016
14	448	001—007	011—017
15	480	001—008	011—017
16	512	001—008	011—018

2.3 Power Bay Configurations

The total number of compute power bays per rack depends on the 48-Vdc connections. If the number of connections is less than or equal to 8, then only one power bay is required in a rack.

A second power bay is required when:

- A fifth C brick is added to a rack
- An eighth brick (of any type except a D brick) is added to a rack.

The following components require a 48-Vdc connection:

- C, I, P, R, and X bricks
- L2 controller

2.3.1 Brick Power Requirements

Table 2-2 Brick Power Requirements

Rack Component	Power Consumption
C Brick (4-processor)	308 Watts
P Brick	225 Watts
R Brick	60 Watts
X Brick	225 Watts
I Brick	190 Watts
L2 Controller	30 Watts
Expansion/JBOD Enclosure	400 Watts
RAID Enclosure	400 Watts
G Brick	2000 Watts

2.3.2 Power Supply Configurations

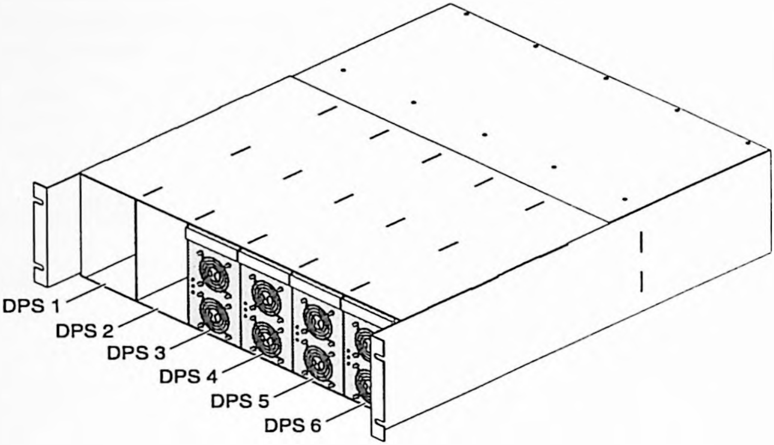


Figure 2-3 Distributed Power Supply Configuration (front)

Table 2-3 Power Bay Configuration

Rack Type	DPS					
	1	2	3	4	5	6
Compute			x	x	x	x
I/O			x	x	x	x

Note: On systems with more than 256 processors, the final DPS location may depend on the phase imbalance of the site's power distribution system.

Table 2-4 PDU Power-phase Configuration

	Single Phase		3 Phase	
Domestic	**013-3074-001**[a]	**013-3310-001**	**013-2896-001**	**013-3311-001**
	2 power drops; 8 poles; 2 circuits; 2 CBs	1 power drop; 4 poles; 1 circuit; 1 CB	60 amp; 1 power drop; 3 poles; 1 circuit; 1 CB	30 amp; 1 power drop; 3 poles; 1 circuit; 1 CB
International	**013-3032-001**	**013-3319-001**	**013-2895-001**	
	2 power drops; 8 poles; 2 circuits; 2 CBs	1 power drops; 4 poles; 1 circuit; 1 CB	32 amp; 1 power drop; 3 poles; 1 circuit; 1 CB	

[a]To ensure power load balance, DPS slots 3 through 6 are populated. There is no power to the auxiliary power strip if you populate bays 1, 2, 3, and 4; one PDU input cord provides power to bays 1, 2, 3, and 4 and the second PDU input cord provides power to bays 5, 6 and the auxiliary outlet on the PDU.

2.4 CPU and Memory Configurations

2.4.1 Memory Configurations

Table 2-5 DIMM Configurations

DIMM Configuration	Part Number	DIMM Capacity	Total Memory Capacity per C Brick	SDRAM Technology
256	030-1018-002	256 MB	512 MB — 2 GB	128 Mbit
512 (premium)[a]	030-1042-003	512 MB	1 GB — 4 GB	128 Mbit
512 (standard)	030-1044-001	512 MB	1 GB — 4 GB	128 Mbit
1024 (premium)	030-1060-003	1 GB	2 GB — 8 GB	256 Mbit

[a]Systems with greater than 128 processors require premium directory memory.

Table 2-6 CPU Slice and Console Subchannel Numbers

CPU	Subchannel
A	0
B	1
C	2
D	3
Console	4

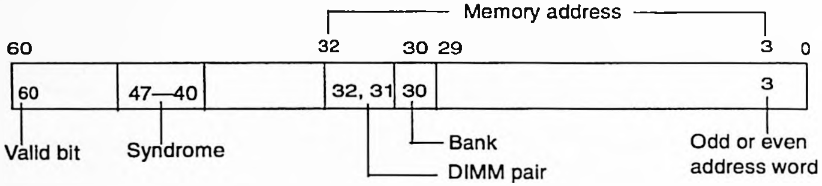


Figure 2-4 Address Bit Layout

To decode error information in the error log, fill out the form at the following URL:

http://wwwcf.americas.sgi.com/PUBLIC/sn1_diags/elog_decode/elog_decode.cgi

You may also break down the hexidecimal memory error and use the data in Table 2-7 or use the DIMMwit program to isolate a failing DIMM. The DIMMwit program is available on the SGI Origin 3000 Server Series Service Information CD-ROM, number CD-SGI3000-1, or from the following URL:

http://servinfo.csd.sgi.com/sgi3000/DIMMwit/index.html

Note: The DIMMwit program isolates to the failing DIMM only if the failing bit is known; otherwise it isolates the failure to a DIMM pair.

Table 2-7 Memory Error Bit Definitions

Bit(s)	Definition
60	Valid bit
61-60	Uncorrectable memory read
53-52	ECC correctable memory read
47-40	Memory syndrome
32-3	Memory address
32-31	DIMM pairs 0—3
30	Bank 0 or 1
3	Determines odd or even word

Table 2-8 Memory Remapping[a]

Address Bit	Bits	DIMM
Even Address (bit 3 = 0)	0 — 7	Odd DIMM
	15 — 8	Even DIMM
	23 — 16	Odd DIMM
	31 — 24	Even DIMM
	39 — 32	Odd DIMM
	47 — 40	Even DIMM
	55 — 48	Odd DIMM
	63 — 56	Odd DIMM
	71 — 64	Even DIMM
Odd Address (bit 3 = 1)	0 — 7	Odd DIMM
	15 — 8	Odd DIMM
	23 — 16	Odd DIMM
	31 — 24	Odd DIMM
	39 — 32	Even DIMM
	47 — 40	Even DIMM
	55 — 48	Even DIMM
	63 — 56	Even DIMM
	71 — 64	Even DIMM

[a] Even DIMM refers to the lower-numbered DIMM and Odd DIMM refers to the higher-numbered DIMM.

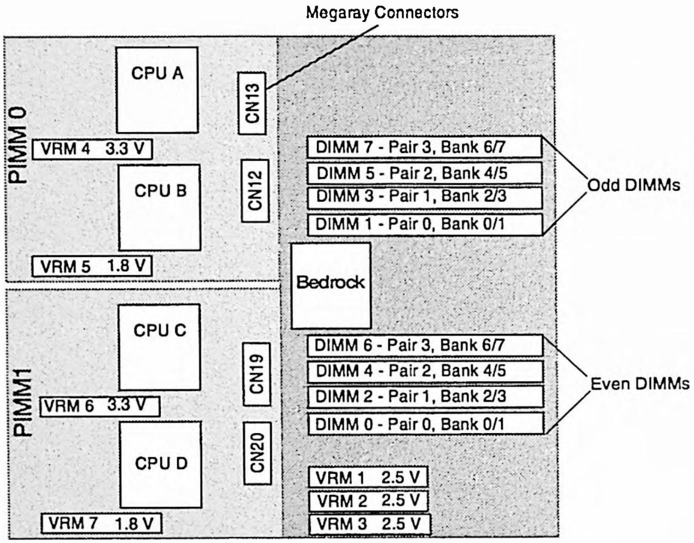


Figure 2-5 C-brick DIMM, PIMM, and VRM Locations

Note: If bank 0 fails, then bank 1 becomes logical bank 0; if bank 0 and bank 1 fail, then bank 2 becomes logical bank 0; if banks 0, 1, and 2 fail, then bank 3 becomes logical bank 0.

2.5 I- and P-brick Configuration and Slot Locations

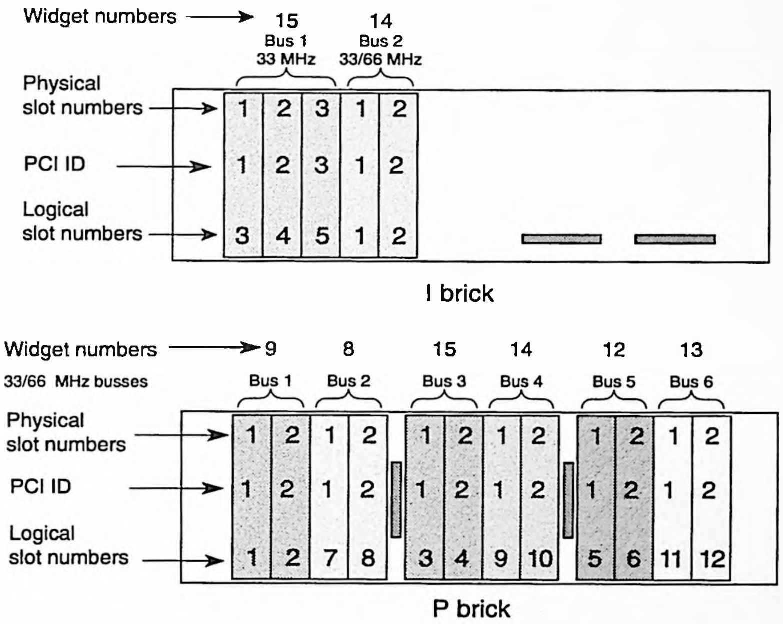


Figure 2-6 P- and I-brick Slot Locations

2.5.1 P- and I-brick VRM Locations

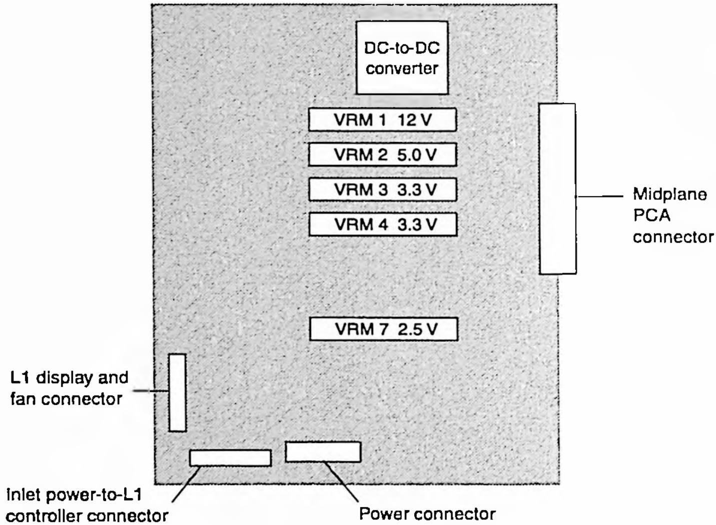


Figure 2-7 I-brick Power Board and VRM Locations

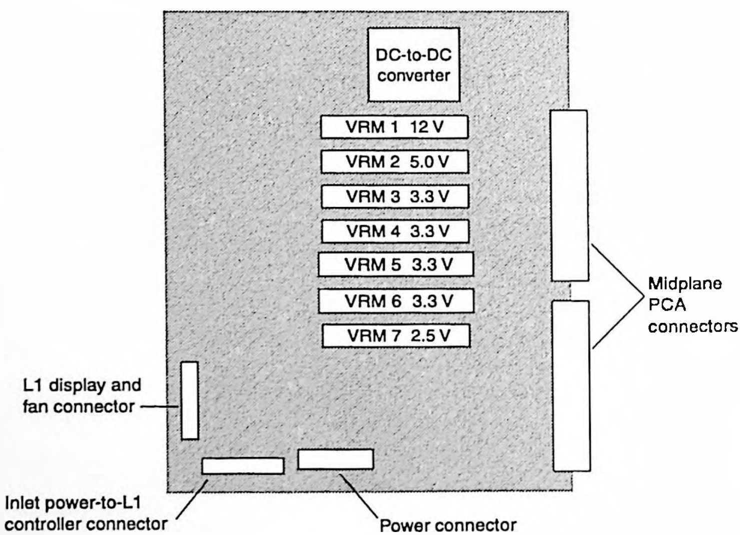


Figure 2-8 P-brick Power Board and VRM Locations

2.5.2 Supported PCI Cards

The URL below shows the appropriate host bus adapters (HBAs) and cables to use in Direct Attach and Switch Attach configurations for various SGI products. Refer to this URL for the up-to-date cable and PCI card information. The minimum IRIX level required to support all PCI cards is IRIX release 6.5.9.

http://sales.corp.sgi.com/products/servers/storage/adapters_cables/index.html

Table 2-9 Supported PCI Cards

Marketing Code for PCI Cards	PCI Card Description	Maximum number of Cards per I-brick 33/66 Mhz	P brick	System tested (untested)	PCI Bus Clock/Data	Peak Bandwidth (per port)
PCI-FC-1PCOP-A (FCA-2)	PCI single channel non-optical Fibre Channel adapter, cable not included (66MHz)	2/2	12	24(84)	66-Mhz/64-bit	100 MB/s
PCI-FC-1POPT-A (FCA-1)	PCI single channel optical Fibre Channel adapter, cable not included (66MHz)	2/2	12	24(84)	66-Mhz/64-bit	100 MB/s
PCI-ATMOC3-1P (ATM-1)	SGI PCI OC3 1-port, MMF ATM NIC; order with SC4-ATMOC3-5.1	0/2	12	8(16)	33-Mhz/64-bit	19.3 MB/s
PCI-ATMOC12-1P (ATM-2)	SGI PCI OC12 1-port, MMF ATM Adapter; order with SC4-ATMOC12_5.x	0/1	6	8(16)	66-Mhz/64-bit	78 MB/s
PCI-GIGENET-C (ETH-2)	1-port copper Gigabit Ethernet	0/1	6	10(30)	66-Mhz/64-bit	125 MB/s

Table 2-9 Supported PCI Cards

Marketing Code for PCI Cards	PCI Card Description	Maximum number of Cards per: I-brick 33/66 Mhz	P brick	System tested (untested)	PCI Bus Clock/Data	Peak Bandwidth (per port)
PCI-GIGENET-C (ETH-1)	SINGLE PORT 1000BASE-SX Gigabit Ethernet adapter; supported in IRIX 6.4	0/1	6	10(30)	66-Mhz/64-bit	125 MB/s
PCI-SCSI-DF-2P (SCS-1)	PCI dual port Ultra SCSI HVD adapter	2/2	12	24(84)	33-Mhz/64-bit	40 MB/s (wide) 20 MB/s (narrow)
PCI-SCSI-LVD-2P (SCS-2)	PCI dual port Ultra SCSI single-ended or Ultra2 SCSI LVD adapter	1/2	12	12(42)	33-MHz/64-bit	80 MB/s (wide) 40 MB/s (narrow)
PCI-AUD-D1000 (AUD-1)	8 Channel Digital Audio PCI Card with BNC AES connector	TBD	TBD	TBD	33-MHz/32-bit	2.88 MB/s
PCI-SER-10002 (SER-1)	PCI serial card	TBD	TBD	TBD	33-MHz/32-bit	.0143 MB/s

2.5.3 PCI Card Configuration Rules

Evenly distribute I/O bandwidth across the I/O bricks (P and I) in the system. To enhance performance, apply the following rules when you populate PCI slots:

- Avoid mixing storage controller cards with networking cards.
- Avoid mixing 33-MHz and 66-MHz cards on the same 66-MHz bus.
- Avoid configuring PCI cards so that the bandwidth of the brick is exceeded (this rule can be violated for capacity configurations).

Note: Refer to the following URL for up-to-date configuration information:
http://wwwcf.americas.sgi.com/PUBLIC/tech_pub/

2.5.4 PCI Card Limits

Each PCI card is given a weighted value as outlined in Table 2-10. The maximum number of a specific type of PCI card within a system is calculated by dividing the aggregate total (a set value) by the weighted value (84/2=42).

Table 2-10 PCI Card Weighted Values

PCI Card	Weighted Value	Aggregate Total	Maximum Number of Cards
FCA-1	1		
FCA-2	1	84	84
SCS-1	1		
SCS-2	2	84	42
ETH-1	1	20	20
ETH-2	1		
ATM-1	1	16	16
ATM-2	1		
AUD-1	1	6	6
SER-1	2	Undefined	Undefined

2.5.5 Adding PCI Cards in a System

Use the flowcharts and the PCI matrix to ensure optimum system bandwidth and performance. The flowcharts help you determine:

1. Determine the PCI slotting mode (Refer to Figure 2-9).
2. Determine where to install cards when the system is in loose-pack mode (Figure 2-10) or close-packed mode (Figure 2-11 and PCI slotting matrix in Table 2-11).

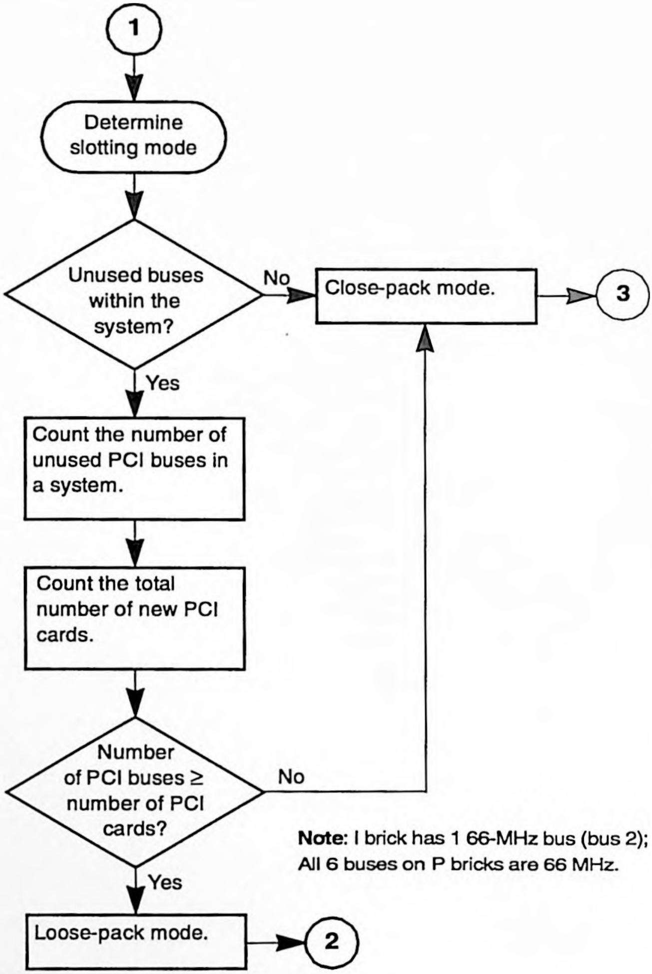


Figure 2-9 PCI Slotting Mode

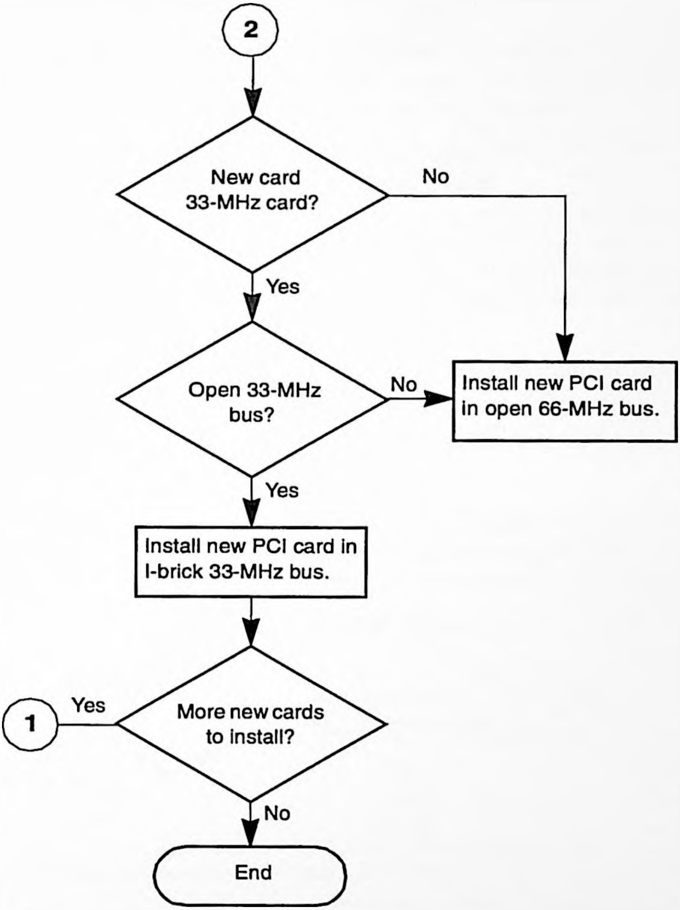


Figure 2-10 Adding PCI Cards to a System in Loose-pack Mode

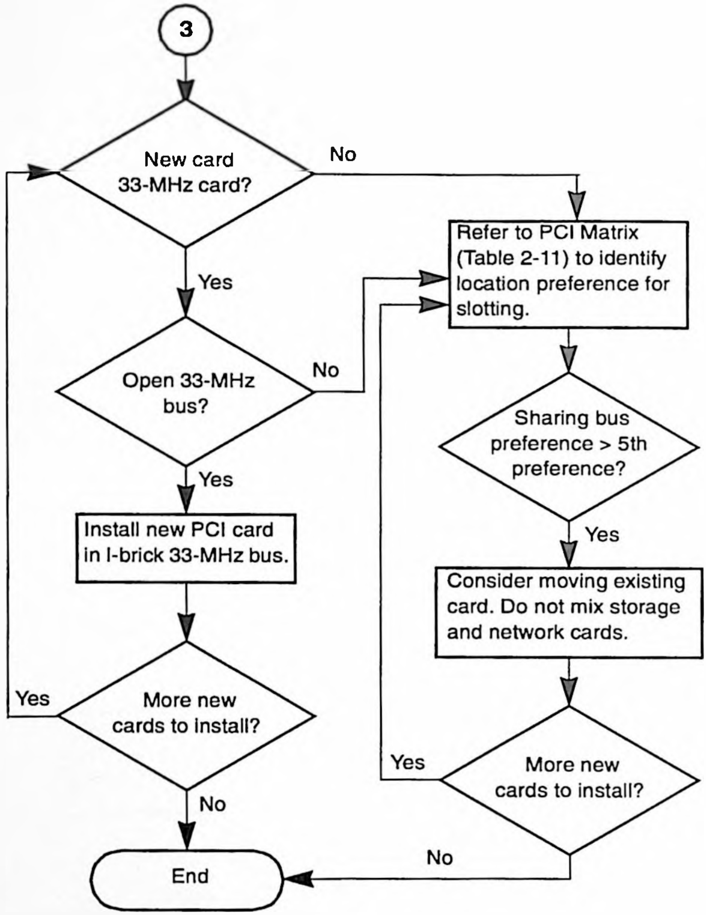


Figure 2-11 Adding PCI Cards to a System in Close-pack Mode

2.5.5.1 Using the PCI Slotting Matrix

Use the PCI slotting matrix to add a PCI card to a system that is in close-packed mode. Refer to Table 2-11 and the following instructions to ensure optimum system bandwidth and performance.

1. Look down column 1 and locate the card type that you want to install in the system.
2. Follow the row across to find a card with which the new card can share a bus. If your system does not have a PCI card with a sharing preference of 5 or less, you may need to move an existing card(s) to another bus. Use the matrix to identify the card with which existing PCI cards can share a bus.
3. Repeat Steps 1 and 2 until all new PCI cards are installed in the system.

   Note: Do not mix storage cards and network cards on the same bus.

If you need to add many cards and require more detail or need to rearrange existing cards for optimum bandwidth, refer to the configuration rules at the following URL:

http://wwwcf.americas.sgi.com/PUBLIC/tech_pub/

Table 2-11 PCI Slotting Matrix

PCI Card		Slotting Preference when Sharing Busses									
Type	Bus Type[a]	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th
ATM-2	66 MHz	SER-1	AUD-1	ATM-1	ATM-2	ETH-1	ETH-2	FCA-1	FCA-2	SCS-1	SCS-2
ATM-1	33 MHz	SER-1	AUD-1	ATM-1	ATM-2	ETH-1	ETH-2	SCS-1	SCS-2	FCA-1	FCA-2
ETH-1	66 MHz	SER-1	AUD-1	ATM-2	ATM-1	ETH-1	ETH-2	FCA-1	FCA-2	SCS-1	SCS-2
ETH-2	66 MHz	SER-1	AUD-1	ATM-2	ATM-1	ETH-1	ETH-2	FCA-1	FCA-2	SCS-1	SCS-2
FCA-1	66 MHz	FCA-1	FCA-2	SCS-1	SCS-2	SER-1	AUD-1	ATM-1	ATM-2	ETH-1	ETH-2
FCA-2	66 MHz	FCA-2	FCA-1	SCS-1	SCS-2	SER-1	AUD-1	ATM-1	ATM-2	ETH-1	ETH-2
SCS-2	33 MHz	SCS-1	SCS-2	FCA-1	FCA-2	SER-1	AUD-1	ATM-1	ATM-2	ETH-1	ETH-2
SCS-1	33 MHz	SCS-1	SCS-2	FCA-1	FCA-2	SER-1	AUD-1	ATM-1	ATM-2	ETH-1	ETH-2
AUD-1	33 MHz	AUD-1	SER-1	SCS-1	SCS-2	ATM-1	ATM-2	ETH-1	ETH-2	FCA-1	FCA-2
SER-1	33 MHz	SER-1	AUD-1	SCS-1	SCS-2	ATM-1	ATM-2	ETH-1	ETH-2	FCA-1	FCA-2

[a] First preference is one card per bus.

2.6 X-brick Configuration and Slot Locations

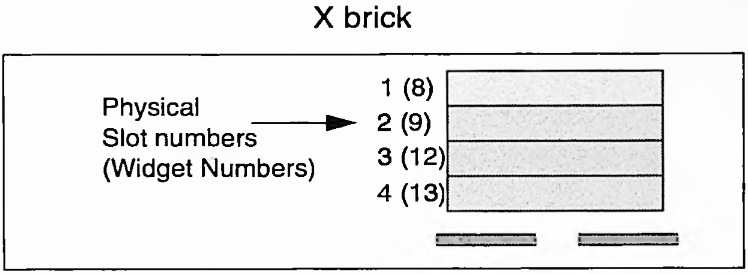


Figure 2-12 X-brick Physical Slot Numbering

2.6.1 X-brick Power Board and VRM Locations

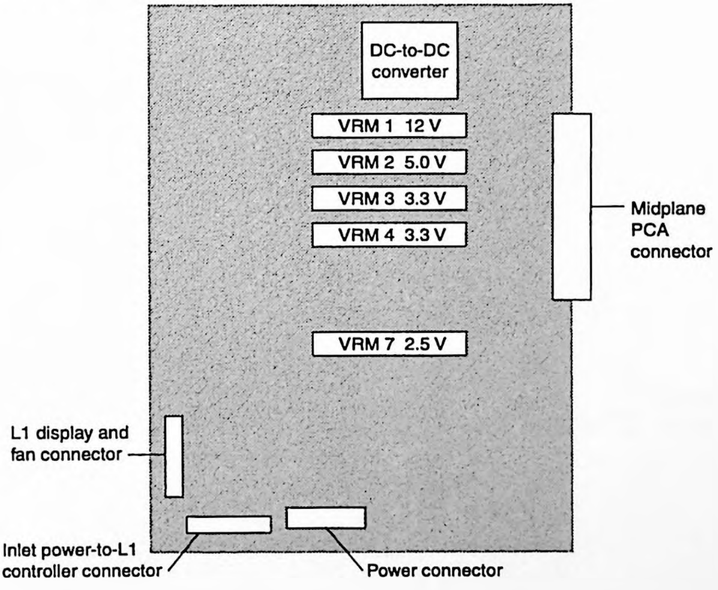


Figure 2-13 X-brick Power Board and VRM Locations

2.6.2 XIO Card Configuration Rule

Evenly distribute I/O bandwidth across the I/O bricks in the system.

2.6.3 XIO Card Limits

Table 2-12 XIO Card System Limits

XIO Card	Maximum number of cards per brick[a]	Maximum number of cards per system
GSN-1	1	4
GSN-2	1	4
DVO-1	4	TBD
DVO-2	4	TBD
FDI-1	4	6
HDV-1	4	TBD
HPI-1	4	16
VME-6	4	4
VME-9	4	4
ATM-3	4	16

[a] If the XIO cards do not draw more than 800 Mbyte/s, maximum cards per brick can be 4; however, these configurations have not been tested.

2.7 D-brick Configuration

Figure 2-14 illustrates the disk enclosure numbering sequence. Use the *sesmgr* command to collect status of all supported Fiber Channel enclosures and change the disk configuration. This IRIX command is a user front end to *sesdaemon*.

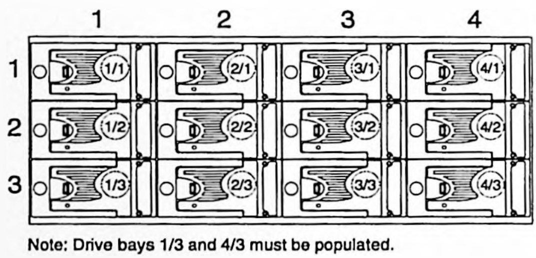


Figure 2-14 Disk Enclosure Numbering Sequence

2.7.1 RAID Drive Configuration

Disk drives are assigned in sequence (refer to Figure 2-15). If a drive bay is not populated, the numbering sequence continues with the next populated bay (refer to Figure 2-16.)

Note: Channel 0/Drive 4 and Channel 1/Drive 5 must be populated.

Ch.0	Ch. 1	Ch.0	Ch. 1
0	0	1	1
2	2	3	3
4	4	5	5

Figure 2-15 D-brick (RAID) Channel Assignments (fully populated D brick)

Ch.0	Ch. 1	Ch.0	Ch. 1
0	0		1
1	2	2	3
3	4	4	5

Figure 2-16 D-brick (RAID) Channel Assignments

2.7.2 JBOD Drive Configuration

Figure 2-17 shows the vault numbers and their corresponding drive ID numbers. From the rear of a D brick (refer to Figure 3-10), the I/O module on the left controls channel 0. SCSI ID numbers are slot dependent; drive ID numbers are fixed and do not change when a drive is removed. The vault number determines the range of SCSI IDs. Vault numbers 0, 8, and 9 are not supported.

	Drive ID			
Vault No.	Ch.0	Ch. 1	Ch.0	Ch. 1
1	109	108	107	106
	105	104	103	102
	101	100	99	98
2	93	92	91	90
	98	88	87	86
	85	84	83	82
3	77	76	75	74
	73	72	71	70
	69	68	67	66
4	61	60	59	58
	57	56	55	54
	53	52	51	50
5	45	44	43	42
	41	40	39	38
	37	36	35	34
6	29	28	27	26
	25	24	23	22
	21	20	19	18
7	13	12	11	10
	9	8	7	6
	5	4	3	2

Figure 2-17 JBOD Drive Configuration

Chapter 3

Cabling and Connections

Subsection	Page Number
DNET Cable Connections	page 3-2
NUMAlink 3 Cabling Guidelines	page 3-2
Xtown2 Cabling Guidelines	page 3-2
Console Connections and Global Master	page 3-3
Brick Connectors	page 3-5
C Brick	page 3-6
R Brick	page 3-8
P Brick	page 3-10
X Brick	page 3-13
I Brick	page 3-16
D Brick (JBOD)	page 3-20
D Brick (RAID)	page 3-21
L2 Controller	page 3-23
Power Bay	page 3-24

3.1 DNET Cable Connections

Figure 3-1 illustrates the labels for all DNET cables.

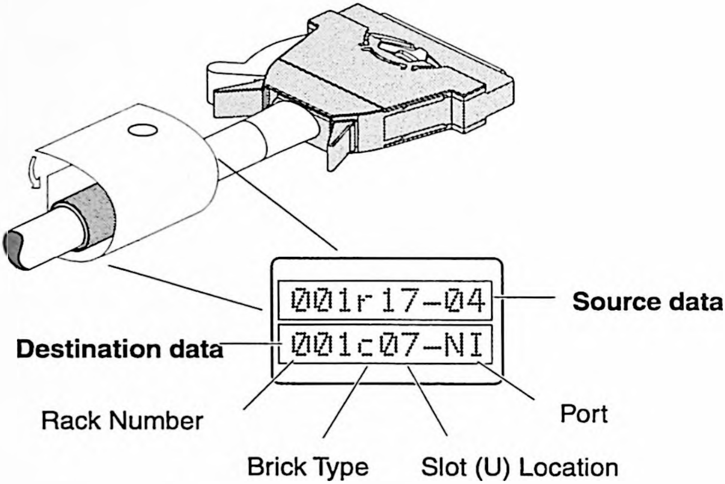


Figure 3-1 DNET Cable Labels

3.1.1 NUMAlink 3 Cabling Guidelines

Refer to the *System Cabling* document, publication number 108-xxxx-002 for detailed information on NUMAlink 3 cable connections and system topology for 3200, 3400, and 3800 configurations. This document is shipped with your system and available online at the following URL: http://servinfo.csd.sgi.com

3.1.2 Xtown2 Cabling Guidelines

- Connect the global master (the first C brick in rack 001) to the I brick in which the system disk resides.
- Connect Xtown2 cables evenly across the hypercube(s).
- Make lateral connections between C and I/O bricks; that is, connect the C brick to the I/O brick that is in the comparable U position in the adjacent rack.
- Route Xtown2 cables horizontally; make any lateral transfers in the compute rack.

3.2 Console Connections and Global Master

Refer to Table 3-1 for the R-brick port and its associative NASID number. Refer to Figure 3-2 and Figure 3-3 for examples of the console connection and global master.

- The global master resides in the lowest numbered C brick that connects to the I brick.
- In 3200 systems, C-brick numbering determines NASID numbering.
- In 3400 and 3800 systems, the R-brick port in which the C brick connects determines NASID numbering. Port 2 is the lowest R-brick port that a C brick can be cabled to.

Table 3-1 3400 and 3800 NASID Numbering

R-brick Port	NASID	R-brick Port	NASID
2/B	0	4/D	2
3/C	1	5/E	3

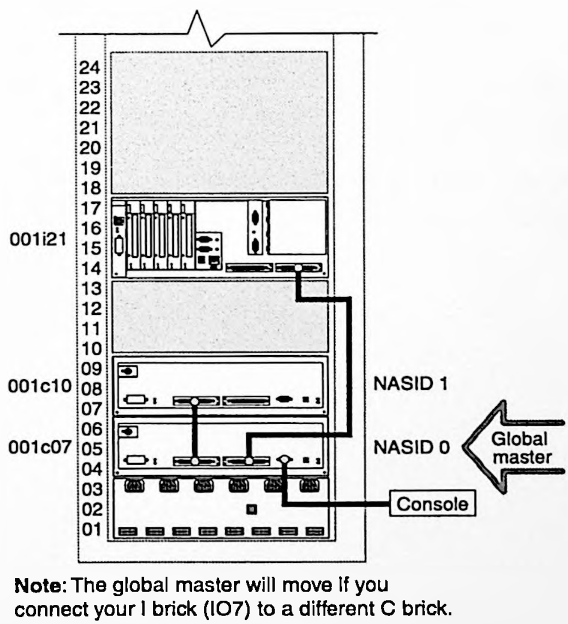
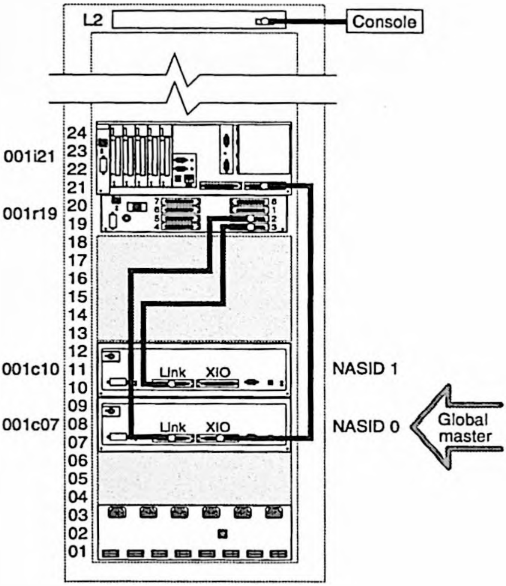


Figure 3-2 3200 Console Connection and Global Master



Notes: The global master will move if you connect your I brick (IO7) to a different C brick. The console connects to the L2 console port or to the L3 Ethernet port.

Figure 3-3 3400 and 3800 Console Connections and Global Master

3.3 Brick Connectors

The following subsections identify the connectors on the rear of each brick. Refer to the *System Cabling* document, publication number 108-xxxx-002 for more detailed information on cabling and connectors.

The following I/O brick figures and tables illustrate screen output for *hinv* and *ioconfig.cfg* file data. These examples identify the correlation between screen output and physical slot and widget (adapter) locations. Figure 3-4 illustrates the components of screen output as it relates to I/O bricks.

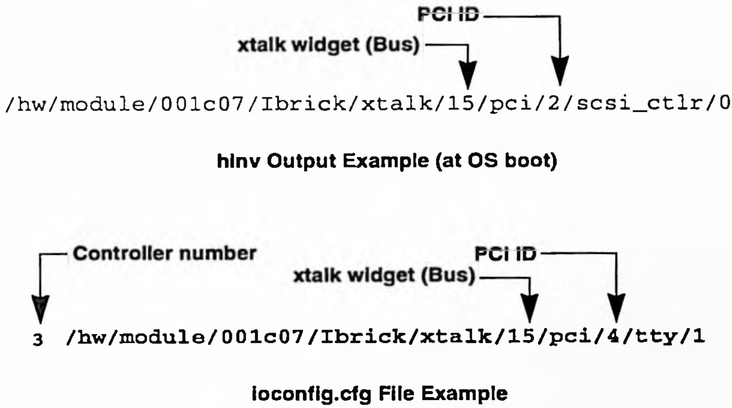


Figure 3-4 `hinv` and `ioconfig.cfg` Screen Format for an I Brick

3.3.1 C Brick

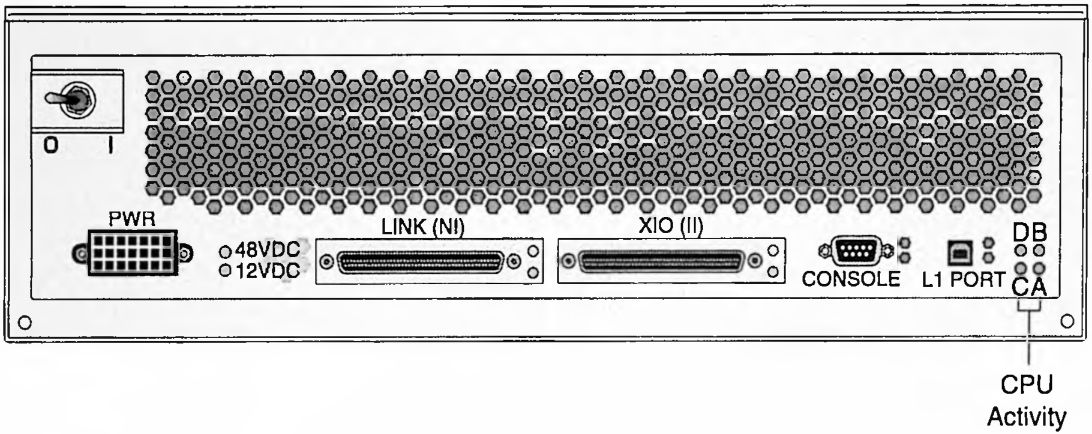


Figure 3-5 C-brick Connectors

Table 3-2 C-brick Connectors

Connector/LED	Type	Internal Connection	External Connection(s)	Notes
Link (NI)	100-pin DNET	Bedrock NI port	C brick: Link (NI) R brick: Links 1—8	C brick to C brick connections use 4 RS-422 wires (two wire pairs where pins 5/6 connect to pins 96/95). C brick to R brick connections use 2 wires for USB signals (R brick pins 5/6 connect to C brick pins 96/95). Two wires are not used (pins 95/6 and 96/5).

Table 3-2 C-brick Connectors **(continued)**

Connector/LED	Type	Internal Connection	External Connection(s)	Notes
XIO (II)	100-pin DNET	Bedrock II port	XIO port 10 or 11 on an I, P, or X bricks	
Console	DB-9		Dumb terminal or L3	
L1 Port	Series A USB	USB Controller	R brick	This port is not used in configurations that do not have an R brick. This port can also be an alternate input port for connecting to a laptop (L3) to perform maintenance activities.
12 Vdc	Green LED	IP35-->Power Board-->DPS-->PDU	--	Illuminated means power is applied to the L1 controller logic.
48 Vdc	Green LED	IP35-->Power Board-->DPS-->PDU	--	Illuminated means power is applied to the brick.
Link (NI) LED XIO (II) LED	Green Yellow		--	Green=good connection to another brick; yellow=software packets being transferred across the link
A, B, C, D	Red	IP35 Motherboard		Heartbeat or CPU activity LED. Blinking=processing; solid=hang; off=no processing and possible failure.

3.3.2 R Brick

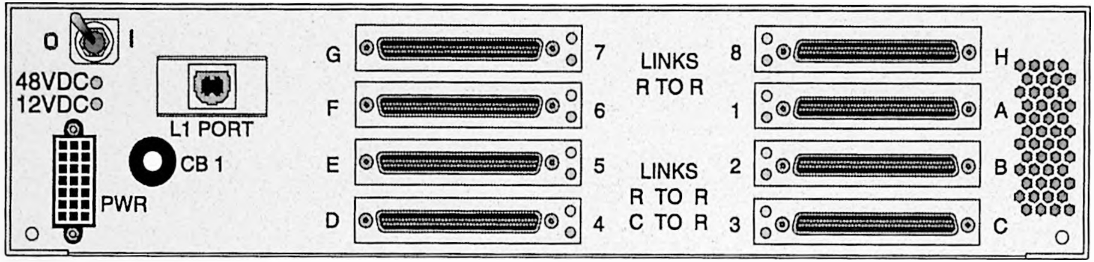


Figure 3-6 R-brick Connectors

Table 3-3 R-brick Connectors

Connector	Type	Internal Connection	External Connection(s)	Notes
L1 Port	Series A USB	L1 USB hub	L2 controller ports 1, 2, 3, or 4	
8	100-pin DNET	Router ASIC Port H	R brick	
1	100-pin DNET	Router ASIC Port A	R brick	
6	100-pin DNET	Router ASIC Port F	R brick	
7	100-pin DNET	Router ASIC Port G	R brick	
2	100-pin DNET	Router ASIC Port B	C or R brick	These ports (C brick-to-R brick connections) use 2 wires for USB signals (R brick pins 5/6 connect to C brick pins 96/95); two wires are not used (pins 95/6 and 96/5)
3	100-pin DNET	Router ASIC Port C	C or R brick	
4	100-pin DNET	Router ASIC Port D	C or R brick	
5	100-pin DNET	Router ASIC Port E	C or R brick	
12 Vdc	Green LED	Power Board--> DPS-->PDU	--	Illuminated means power is applied to the L1 controller logic
48 Vdc	Green LED	Power Board--> DPS-->PDU	--	Illuminated means power is applied to the brick
Links R to R LEDs; Links C to R LEDs	Green Yellow	--	--	Green=a good connection to another brick; yellow=software packets being transferred across the link

3.3.3 P Brick

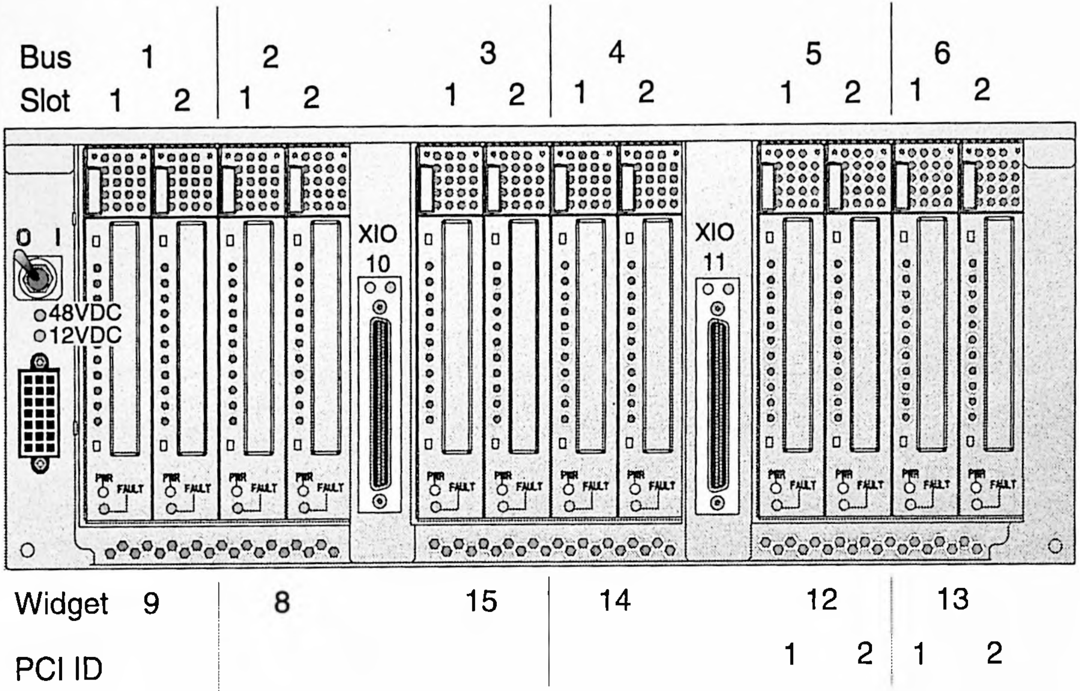


Figure 3-7 P-brick Connectors

Table 3-4 Screen Output Showing P-brick Configuration

`$ cd /etc/ioconfig.conf`	`$ hinv -g`
`7 /hw/module/001c10/Pbrick/xtalk/12/pci/1/scsi_ctlr/0`	`/hw/module/001c10/Pbrick/xtalk/12/scsi_ctlr/0/target`

Table 3-5 P-brick Connectors

Connector/LED		Xtalk Widget	PCI ID	X-bridge Data Path/Internal Connection(s)	Connector Type	External Connection(s)	Notes
Bus1	Slot 1 Slot 2	9	1	X bridge U1 Port F-->Port 8-->U0 port 9	184-pin PC	Depends on PCI cards	PCI connector on the motherboard
Bus 2	Slot 1 Slot 2	8	2	X bridge U1 Port E-->Port D--> U0 8	184-pin PC	Depends on PCI cards	PCI connector on the motherboard
Bus 3	Slot 1 Slot 2	15	1	X bridge U0 Port F	184-pin PC	Depends on PCI cards	PCI connector on the motherboard
Bus 4	Slot 1 Slot 2	14	2	X bridge U0 Port E	184-pin PC	Depends on PCI cards	PCI connector on the motherboard
Bus 5	Slot 1 Slot 2	12	1	X bridge U2 Port F-->Port 8-->U0 Port C	184-pin PC	Depends on PCI cards	PCI connector on the motherboard

Table 3-5 P-brick Connectors **(continued)**

Connector/LED		Xtalk Widget	PCI ID	X-bridge Data Path/Internal Connection(s)	Connector Type	External Connection(s)	Notes
Bus 6	Slot 1 Slot 2	13	2	X bridge U2 Port E-->Port D-->U0 Port D	184-pin PC	Depends on PCI cards	PCI connector on the motherboard
XIO 10		10	--	X bridge U0 Port A	100-pin DNET	C-brick XIO port	
XIO 11		11	--	X bridge U0 Port B	100-pin DNET	C-brick XIO port	
48-Vdc LED		--	--	Power Board-->DPS-->PDU	Green	--	Illuminated means power is applied to the brick
12-Vdc LED		--	--	Power Board--> DPS-->PDU	Green	--	Illuminated means power is applied to the L1 controller logic
XIO 10 LED XIO 11 LED		--	--		Green Yellow	--	Green=good connection to another brick; yellow =software packets being transferred across the link
PWR LED (PCI Carrier)		--	--	PCI motherboard-->power board	Green LED	--	Power on
Fault LED (PCI Carrier)		--	--	PCI motherboard-->power board	Yellow LED	--	Fault occurred

3.3.4 X Brick

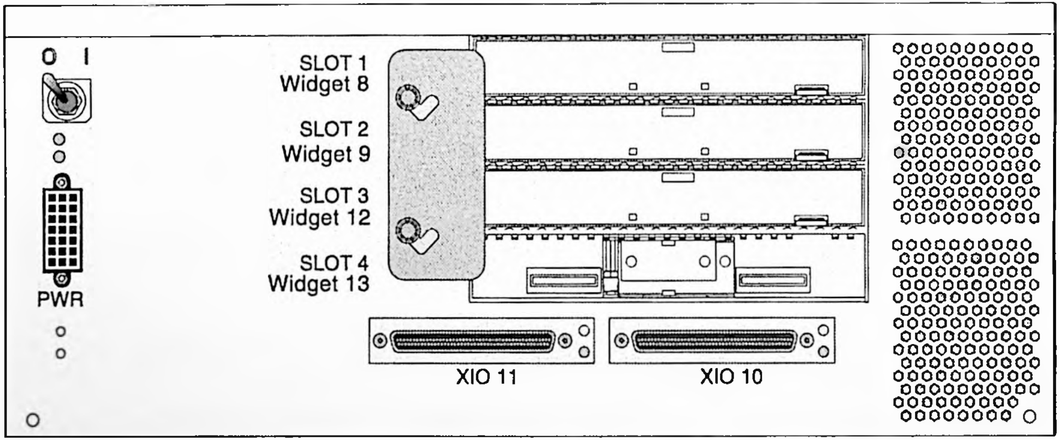


Figure 3-8 X-brick Connectors

Table 3-6 Screen Output Showing X-brick Configuration

`$ cd /etc/ioconfig.conf`	`$ hinv -g`
`7 /hw/module/001c10/Xbrick/xtalk/12`	`/hw/module/001c10/Xbrick/xtalk/12`

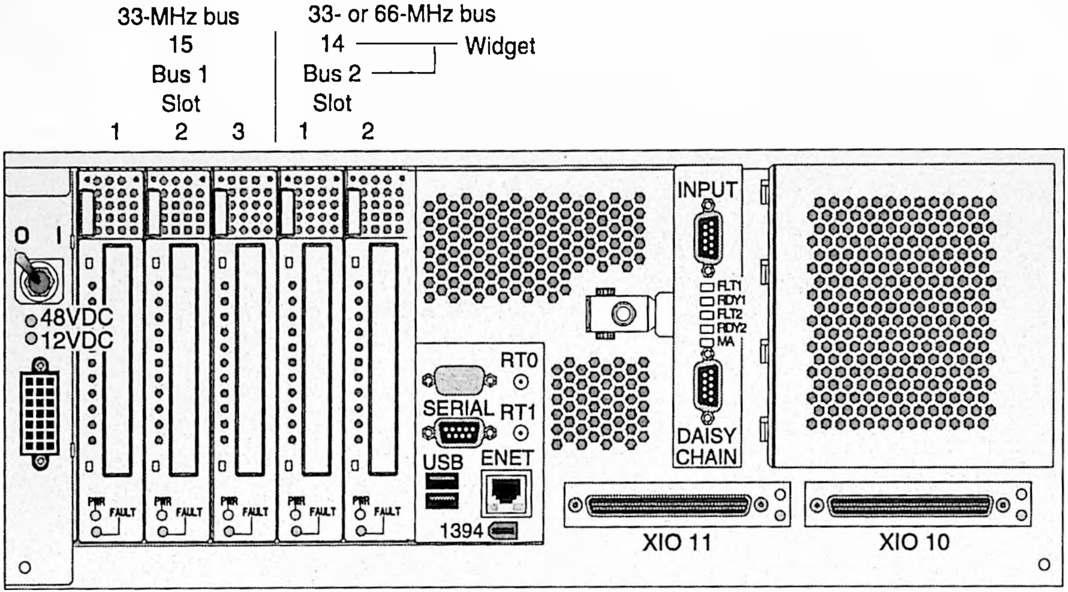
Table 3-7 X-brick Connectors

Connector/LED	Xtalk Widget	PCI ID	Xbridge Data Path/Internal Connection(s)	Connector Type	External Connection(s)	Notes
Slot 1	8		Xbridge Port 8	100-pin (4x25) PC	Depends on XIO cards	Crosstown interface XIO PCA connector
Slot 2	9		Xbridge Port 9	100-pin (4x25) PC	Depends on XIO cards	Crosstown interface XIO PCA connector
Slot 3	12		Xbridge Port C	100-pin (4x25) PC	Depends on XIO cards	Crosstown interface XIO PCA connector
Slot 4	13		Xbridge Port D	100-pin (4x25) PC	Depends on XIO cards	Crosstown interface XIO PCA connector
XIO 10	10		Xbridge Port A	100-pin DNET	C-brick XIO port	
XIO 11	11		Xbridge Port B	100-pin DNET	C-brick XIO port	
48-Vdc LED	--	--	Power Board-->DPS-->PDU	Green	--	Illuminated means power is applied to the brick
12-Vdc LED	--	--	Power Board--> DPS-->PDU	Green	--	Illuminated means power is applied to the L1 controller logic.

Table 3-7 X-brick Connectors **(continued)**

Connector/LED	Xtalk Widget	PCI ID	Xbridge Data Path/Internal Connection(s)	Connector Type	External Connection(s)	Notes
XIO 10 LED XIO 11 LED	--	--		Green Yellow	--	Green=good connection to another brick; yellow=software packets being transferred across the link

3.3.5 I Brick



Note: Bus 1 slots 1, 2, and 3 are for 33-MHz cards only. Bus 2 slots 1 and 2 can handle 33- or 66-MHz PCI cards.

Figure 3-9 I-brick Connectors

Table 3-8 Screen Output Showing I-brick Configuration

`$ cd /etc/ioconfig.conf`	`$ hinv -g`
`3 /hw/module/001c07/Ibrick/xtalk/15/pci/4/tty/1`	`/hw/module/001c07/Ibrick/xtalk/15/pci/4/ioc30`

Table 3-9 I-brick Connectors

Connector/LED		Xtalk Widget	PCI ID	Xbridge Data Path/Internal Connection	Connector Type	External Connection	Notes
Bus 1	Slot 1	15	1	X bridge (Port F)	184-pin PC	Depends on PCI cards	PCI connector on the 12-slot I/O printed circuit assembly
	Slot 2		2				
	Slot 3		3				
Bus 2	Slot 1	14	1	X bridge (Port E)	184-pin PC	Depends on PCI cards	PCI connector on the 12-slot I/O printed circuit assembly
	Slot 2		2				
Serial		15	0	X bridge (U715-->U714-->U701-->Port F)	RS-232 (DB-9)	Console or dumb terminal	
USB 1		15	5	X bridge (U702 Port 1-->Port F)	USB	USB hub and peripherals	
USB 2		15	5	X bridge (U702 Port 2-->Port F)	USB	USB hub and peripherals	

Table 3-9 I-brick Connectors **(continued)**

Connector/LED	Xtalk Widget	PCI ID	Xbridge Data Path/Internal Connection	Connector Type	External Connection	Notes
1394	15	6	X bridge (U707--> U706-->Port F)			FireWire; serial-bus protocol connection for digital media, storage, networking, and other applications
RTO	15	4	X bridge (U701-->Port F)	Stereo Jack		Real-time interrupt out; used primarily with graphics
RTI	15	4	X bridge (U701-->Port F)	Stereo Jack		Real-time interrupt in; used primarily with graphics
ENET	15	4	X bridge (U708-->U701->Port F)		Network	Connects to the Ethernet hub
XIO 11	11		X bridge Port B	100-pin DNET	C-brick XIO port.	
XIO 10	10		X bridge Port A	100-pin DNET	C-brick XIO port.	
Input				DB-9	Fibre Channel PCI card	
Daisy Chain				DB-9	Connects to another drive	Can daisy chain to I or D bricks, or external drives
48-Vdc LED	--	--	Power Board-->DPS-->PDU	Green	--	Illuminated means power is applied to the brick

Table 3-9 I-brick Connectors **(continued)**

Connector/LED	Xtalk Widget	PCI ID	Xbridge Data Path/Internal Connection	Connector Type	External Connection	Notes
12-Vdc LED	--	--	Power Board-->DPS--> PDU	Green	--	Illuminated means power is applied to the L1 controller logic
XIO 10 LED XIO 11 LED	--	--		Green Yellow	--	Green=good connection to another brick; yellow= software packets being transferred across the link
PWR LED (PCI Carrier)	--	--	PCI motherboard--> power board	Green LED	--	
Fault LED (PCI Carrier)	--	--	PCI motherboard--> power board	Yellow LED	--	
ENET LED	--	--	PCI motherboard	Green Yellow	--	

3.3.6 D Brick (JBOD)

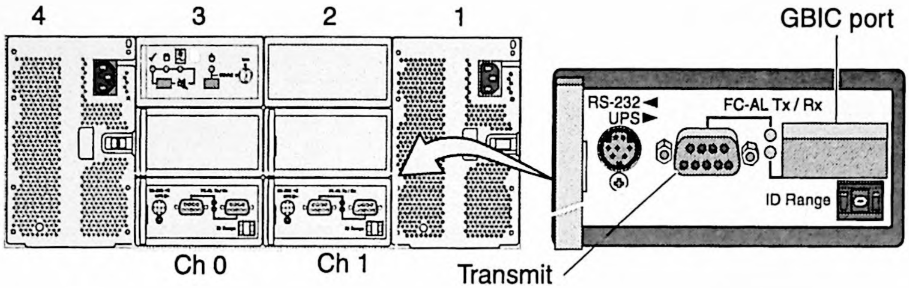


Figure 3-10 D-brick Connectors

Table 3-10 D-brick Connectors (JBOD)

Connector	Connector Type/LED	External Connection(s)	Notes
RS-232/UPS	RS-232		Not used
FC-AL Transmit (Tx) and LED	RS-232; Green/amber LED	Host bus adapter (HBA) in an P or I brick	Green LED=good FC-AL connection; amber LED=problem with connection
GBIC port (RX) and LED	Green/amber LED	Another D brick (Rx port) in the rack configuration	Green LED=good FC-AL connection; amber LED=problem with connection; can be daisy chained to another brick but must set jumper on I/O bulkhead
ID Range	--	--	ID range starts at 1; valid ranges are 1—7

3.3.7 D Brick (RAID)

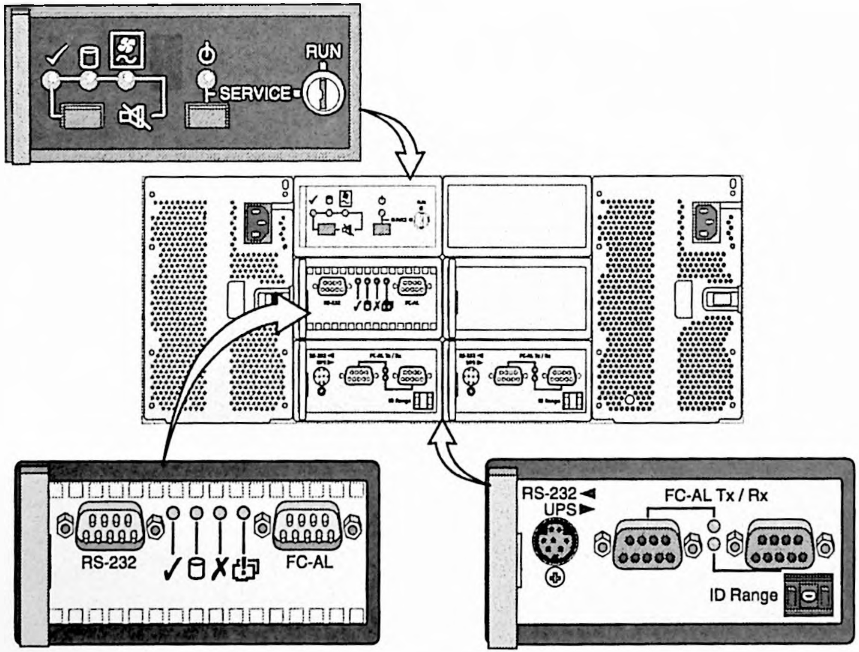


Figure 3-11 D-brick Connectors (RAID)

Table 3-11 D-brick Connectors (RAID)

Connector	Connector Type	External Connection(s)	Notes
RS-232/UPS	RS-232		Not used
FC-AL Transmit (Tx)	RS-232	Another D brick (Rx port) in the rack configuration	Green LED=good FC-AL connection; amber LED=problem with connection; can be daisy chained to another brick but must set jumper on I/O bulkhead
FC-AL Receive (Rx)	RS-232	QLogic PCI card in P or I brick	Green LED=good FC-AL connection; amber LED=problem with connection
ID Range	--	--	ID range starts at 0; valid range is 0—9

3.3.8 L2 Controller

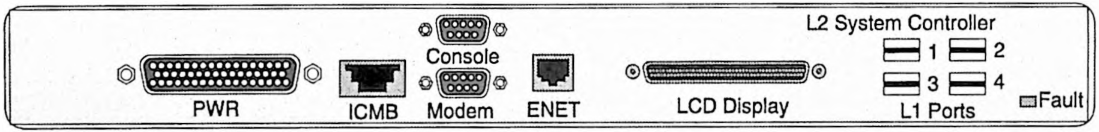


Figure 3-12 L2 Controller Connectors

Table 3-12 L2 Controller Connectors

Connector	Connector Type	Internal Connection	External Connection(s)	Notes
PWR			Power bay	
ICMB	Molex	RS-485 controller --> processor	D brick	Not used
Console	DB-9	RS-232 controller-->processor	Dumb terminal	
Modem	DB-9	RS-232	External modem	Only used when a Cisco router (ISDN) is not used
Ethernet	RJ45	100BT controller	L3, Ethernet hub, or Cisco router	
LCD Display		RS-422 controller --> processor and LCD controller	L2 display	
L1 Ports	USB	USB controller -->PCI Bridge	C, R or USB hub	G bricks may use the USB hub to connect through the L1 ports

3.3.9 Power Bay

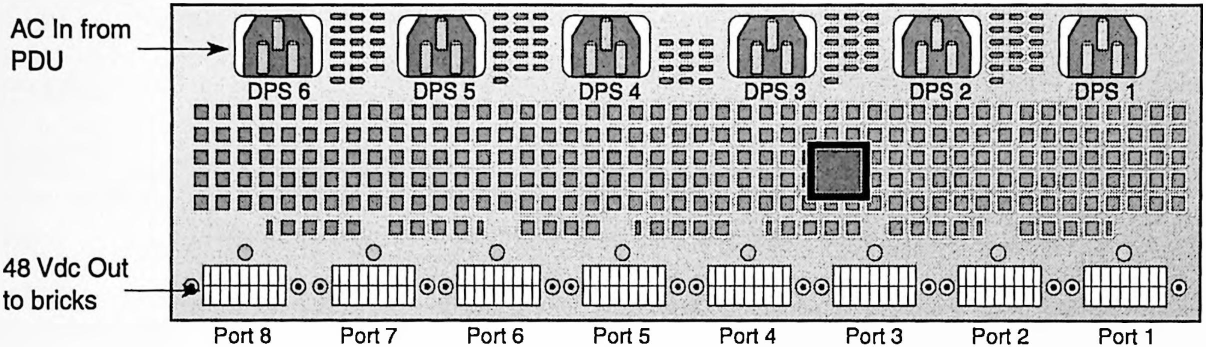


Figure 3-13 Power Bay Connectors

Table 3-13 Power Bay Connectors

Connector	Connector Type	Internal Connection	External Connection(s)	Notes
AC Input DPS 6—DPS1	--	--	PDU	
48-Vdc Output Port 8—Port 1	--	--	Power connector on the bricks and the L2 controller	L2 controller connection requires a 3-meter power cable and the L2 PCA power adapter

Table 3-14 Power Supply LED States

Power Supply Condition	Power (Green)	PFAIL (Amber)	Fail (Amber)
AC voltage not applied to all power supplies	Off	Off	Off
AC voltage not applied to this power supply	Off	Off	Off
AC voltage present; standby voltage on	Blinking	Off	Off
Power supply DC outputs on	On	Off	Off
Power supply failure	Off	Off	On
Current limit reached on 48-Vdc output	On	Off	Blinking
Predictive failure	On	Blinking	Off

Chapter 4

Commands

Subsection	Page Number
L1 Commands	page 4-2
Virtual Debug Switch Settings	page 4-13
L2 Commands	page 4-15
PROM Monitor Commands	page 4-20
POD Commands	page 4-23

4.1 L1 Commands

Table 4-1 Connecting to the L1

	Serial Connection	USB Connection (Emulator)	Ethernet Connection
1.	cu -1 ttyS0 -s 38400	`/stand/sysco/bin`	`/stand/sysco/bin`
2.	Enter	`./l2`	`./l2term`
3.	*r.s command*	*r.s command*	*r.s command*

Note: Use the *r.s* option to target a specific L1 controller from the L2. You may use this option with any L1 command. Type `ctrl t` at the L1 prompt to return to the L2 prompt.

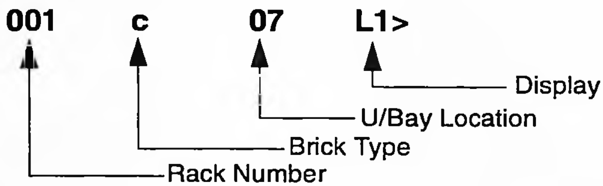


Figure 4-1 L1 Prompt

Table 4-2 L1 Commands (Version 1.4.1)

Command	Description
`autopower|apwr`	Displays current auto powerup state; this is only valid when the system does not contain an L2 controller
`autopower|apwr on|off`	Enables or disables auto power up (for systems without an L2 controller) on L1 power up
`autopower|apwr abort`	Aborts auto power on
`bedrock|bdrck`	Gets L1<-->Bedrock protocol
`bedrock|bdrck ppp|raw`	Sets L1<-->Bedrock protocol to PPP or RAW mode
`bedrock|bdrck test on|off`	Bedrock|bdrck test on or off
`bedrock|bdrck test on`	enabled simulated bedrock output (loop & subchannel)

Table 4-2 L1 Commands (Version 1.4.1)

Command	Description
`brick`	Gets brick information. Need to reboot L1 for brick commands to take effect
`brick type` *<str>*	Sets brick type; *<str>* is one of c, r, p, x, i, ip35, ip37, or g
`brick ptype` *<str>*	Sets brick pseudo-type; *<str>* is one of c, r, p, x, i, ip35, ip37, or g
`brick rack` *<decimal exp>*	Sets brick rack number
`brick rackslot|rs` *<decimal exp> <decimal exp>*	Sets brick rack and slot
`brick slot|bay|upos` *<decimal exp>*	Sets brick slot
`brick partition|part none`	Clears brick partition
`brick partition|part` *<exp>*	Sets brick partition
`config|cfg`	Shows brick L1 configuration
`config|cfg v|verbose`	Shows brick L1 configuration (verbose)
`config|cfg reset`	Resets L1 configuration
`ctc` *<command>*	Sends command to connected C brick
`ctci` *<cmd>*	Sends command to connected C brick's I/O brick
`cti`	Sends command to connected I/O brick
`date`	Displays current L1 date and time
`date <str>`	Sets L1 date and time (date set MMDDhhmm[[CC]YY][.ss])
`date tz`	Displays current time-zone settings
`date tz <str>`	Sets UNIX time-zone string (ex: date tz *CST6CDT*)
`debug`	Gets virtual debug switches
`debug` *<exp>*	Sets virtual debug switch settings (debug 0x0018 sets switches in hex 18; debug 24 sets switches to hex 18 decimal 24)
`display`	Shows L1 display status
`display|dsp` *<exp> <cmd>*	Displays text on the L1 display; *<exp>* is 1 or 2
`display|dsp power|pwr on`	Turns power LED on
`display|dsp attention|attn on`	Turns attention LED on

Table 4-2 L1 Commands (Version 1.4.1)

Command	**Description**
display|dsp failure|fail on	Turns failure LED on
display|dsp power|pwr off	Turns power LED off
display|dsp attention|attn off	Turns attention LED off
display|dsp failure|fail off	Turns failure LED off
env	Gets environmental monitoring status
env altitude high|low	Sets high|low altitude >|<= 5000 ft./1500m temperature limits
env check	Gets a brief environmental monitoring status
env [on|off]	Enables or disables environmental monitoring at L1 boot
env reset|rst	Resets all current warnings and faults
env test warning	Generates an environmental warning test
env temp on|off	Turns normal temperature monitoring on or off
env test fault	Generates an environmental fault test
env test tmpadv	Generates an environmental advisory temperature test
env test tmpcrit	Generates an environmental critical temperature test
eeprom	Shows brick eeprom data
eeprom *<exp>* *<exp>* *<exp>*	Shows brick eeprom data at *<eeprom>* *<offset>* *<length>*
fan	Shows fan status; 8 is considered normal speed. Fan speed commands do not apply to R bricks
fan [on|off]	Turns fan(s) on or off
fan speed	Shows fan speed status
fan speed *<exp>*	Gets fan speed, <exp> is 0....31. Values greater than 31 are considered 0 and power the fan off
flash	
flash default [a|b]	Sets image A or B as the default flash image
flash default reset	Determines default image at boot time
flash default current	Sets current image as the default flash image

Table 4-2 L1 Commands (Version 1.4.1)

<table>
<tr><th>Command</th><th>Description</th></tr>
<tr><td>flash default old</td><td>Sets older image as the default flash image</td></tr>
<tr><td>flash default new</td><td>Sets newer image as the default flash image</td></tr>
<tr><td>flash remove [a|b]</td><td>Erases (invalidates) flash image A or B. This command is currently disabled.</td></tr>
<tr><td>flash serial [a|b]</td><td>Updates flash image A or B via the SMP port</td></tr>
<tr><td>flash serial boot</td><td>Updates flash boot image via the SMP port. This command is currently disabled.</td></tr>
<tr><td>flash status</td><td>Displays the status of flash images</td></tr>
<tr><td>flash ctc|cti|ctci|all</td><td>Updates the L1s flash with default local image</td></tr>
<tr><td>flash ctc|cti|ctci|all |a|b</td><td>Updates the L1s flash with a specified local image</td></tr>
<tr><td>flash util erase all</td><td>Erases entire flash</td></tr>
<tr><td>flash util erase <exp> <exp></td><td>Erases flash <flash offset> <len></td></tr>
<tr><td>flash util program <exp> <str></td><td>Programs flash <flash offset> <string data></td></tr>
<tr><td>flash util test</td><td>Tests flash erase/program</td></tr>
<tr><td>flash util test <exp></td><td>Tests flash erase/program, <exp> passes</td></tr>
<tr><td>help|hlp</td><td>Gets list of commands</td></tr>
<tr><td>help|hlp <str></td><td>Gets help on single command</td></tr>
<tr><td>history|hist</td><td>Shows SMP command history</td></tr>
<tr><td>ioport|ioprt</td><td>Gets I/O port speeds</td></tr>
<tr><td>ioport|ioprt 400</td><td>Sets I/O port A and B to 400 MHz</td></tr>
<tr><td>ioport|ioprt 600</td><td>Sets I/O port A and B to 600 MHz</td></tr>
<tr><td>ioport|ioprt [a|b] [400|600]</td><td>Sets I/O port A or B to 400 or 600 MHz</td></tr>
<tr><td>ioport|ioprt clksrc [a|b]</td><td>Sets UST clock source to Port A or B</td></tr>
<tr><td>istat</td><td></td></tr>
<tr><td>istat memory|m</td><td>Gets L1 memory status</td></tr>
<tr><td>istat queues|q</td><td>Gets L1 queue status</td></tr>
<tr><td>istat tasks|t</td><td>Gets L1 task status</td></tr>
<tr><td>istat pmalloc|p</td><td>Gets L1 task status</td></tr>
<tr><td>l1</td><td>Engages L1 command processor indefinitely</td></tr>
</table>

Table 4-2 L1 Commands (Version 1.4.1)

Command	Description
l1dbg	Gets L1 debug settings
l1dbg bedrock|bdk on|off	Sets Bedrock communication debugging on or off
l1dbg irtr on|off	Sets IRouter debugging on or off
l1dbg env on|off	Sets Environmental debugging on | off
l1dbg port on|off	Sets Port interrupt debugging on or off
l1dbg i2c on|off	Sets I^2C interrupt on or off
l1dbg margin|mgn on|off	Turns L1 voltage margin debugging on or off
l1dbg qsusp <*exp*> <*exp*>	Sets queue suspend time for req/rsp and evt
leds	Shows LED status and displays a description of the LED value; this command samples the LED value multiple times and reports all unique values
log	Dumps log
log reset	Inserts entry into log
log <*cmd*>	Inserts entry into log
margin|mgn	Gets all voltages margin status
margin|mgn default	Sets all margins to ROM defaults
margin|mgn [low|norm|high]	Sets all voltages to low, normal, or high margins
margin|mgn gbrick default	Sets all G-brick voltages to ROM defaults
margin|mgn gbrick low|norm|high	Sets all G-brick voltages to low, normal, or high margins
margin|mgn 1.5 default	Sets 1.5 V to ROM default voltage
margin|mgn 1.5 [low|norm|high]	Sets 1.5 V to low, normal, or high margin
margin|mgn 1.5 <*exp*>	Sets 1.5 V margin, <*exp*> 0....255
margin|mgn 1.8 default	Sets 1.8 V to ROM default voltage
margin|mgn 1.8 [low|norm|high]	Sets1.8 V to low, normal, or high margin
margin|mgn 1.8 <*exp*>	Sets 1.8 V margin, <*exp*> 0....255
margin|mgn 2.5 default	Sets 2.5 V to ROM default voltage
margin|mgn 2.5 [low|norm|high]	Sets 2.5 V to low, normal, or high margin

Table 4-2 L1 Commands (Version 1.4.1)

<table>
<tr><th>Command</th><th>Description</th></tr>
<tr><td>margin|mgn 2.5 <exp></td><td>Sets 2.5 V margin, <exp> 0....255</td></tr>
<tr><td>margin|mgn 3.3 default</td><td>Sets 3.3 V to ROM default voltage</td></tr>
<tr><td>margin|mgn 3.3 [low|norm|high]</td><td>Sets 3.3 V to low, normal, or high margin</td></tr>
<tr><td>margin|mgn 3.3 <exp></td><td>Sets 3.3 V margin, <exp> 0....255</td></tr>
<tr><td>margin|mgn 5 default</td><td>Sets 5 V to ROM default voltage</td></tr>
<tr><td>margin|mgn 5 [low|norm|high]</td><td>Sets 5 V to low, normal, or high margin</td></tr>
<tr><td>margin|mgn 5 <exp></td><td>Sets 5 V margin, <exp> 0....255</td></tr>
<tr><td>margin|mgn VTERM default</td><td>Sets VTERM to ROM default voltage</td></tr>
<tr><td>margin|mgn VTERM low|norm|high</td><td>Sets VTERM to low, normal, or high margin</td></tr>
<tr><td>margin|mgn dimm default</td><td>Sets DIMM to ROM default voltage</td></tr>
<tr><td>margin|mgn dimm [low|norm|high]</td><td>Sets DIMM to low, normal, or high margin</td></tr>
<tr><td>margin|mgn dimm <exp></td><td>Sets DIMM margin, <exp> 0....255</td></tr>
<tr><td>margin|mgn dimm v|voltage <str></td><td>Sets DIMM margin based on specified target voltage <exp>; this allows a C bricks DIMM VRM's to be margined to a specific voltage without using the low | normal | high settings</td></tr>
<tr><td>margin|mgn 1.5 default</td><td>Sets 1.5 V to ROM default voltage</td></tr>
<tr><td>margin|mgn 1.5 [low|norm|high]</td><td>Sets 1.5 V to low, normal, or high margin</td></tr>
<tr><td>margin|mgn 1.5 <exp></td><td>Sets 1.5 V margin, <exp> 0....255</td></tr>
<tr><td>margin|mgn 1.5p0 default</td><td>Sets PIMM0 1.5 V ROM default voltage</td></tr>
<tr><td>margin|mgn 1.5p0 [low|norm|high]</td><td>Sets PIMM0 1.5 V to low, normal, or high margin</td></tr>
<tr><td>margin|mgn 1.5p0 <exp></td><td>Sets PIMM0 1.5 V margin, <exp> 0....255</td></tr>
<tr><td>margin|mgn 1.5p1 default</td><td>Sets PIMM1 1.5 V ROM default voltage</td></tr>
<tr><td>margin|mgn 1.5p1 [low|norm|high]</td><td>Sets PIMM1 1. 5 V to low, normal, or high margin</td></tr>
<tr><td>margin|mgn 1.5p1 <exp></td><td>Sets PIMM1 1.5 V margin, <exp> 0....255</td></tr>
<tr><td>margin|mgn p0cpu default</td><td>Sets PIMM0 CPU to ROM default voltage</td></tr>
</table>

Table 4-2 L1 Commands (Version 1.4.1)

<table>
<tr><th>Command</th><th>Description</th></tr>
<tr><td>margin|mgn p0cpu [low|norm|high]</td><td>Sets PIMM0 CPU to low, normal, or high margin</td></tr>
<tr><td>margin|mgn p1cpu default</td><td>Sets PIMM1 CPU margin, <<i>exp</i>> 0....255</td></tr>
<tr><td>margin|mgn p1cpu [low|norm|high]</td><td>Sets PIMM1 CPU to low, normal, or high margin</td></tr>
<tr><td>margin|mgn p1cpu <<i>exp</i>></td><td>Sets PIMM1 CPU margin, <<i>exp</i>> 0....255</td></tr>
<tr><td>margin|mgn p0sram</td><td>Sets PIMM0 SRAM to ROM default voltage</td></tr>
<tr><td>margin|mgn p0sram low|norm|high</td><td>Sets PIMM0 SRAM to low, normal, or high margin</td></tr>
<tr><td>margin|mgn p0sram <exp></td><td>Sets PIMM0 SRAM margin, <exp> 0.255</td></tr>
<tr><td>margin|mgn p1sram default</td><td>Sets PIMM1 SRAM to ROM default voltage</td></tr>
<tr><td>margin|mgn p1sram low</td><td>Sets PIMM1 SRAM to low, normal, or high margin</td></tr>
<tr><td>margin|mgn p1sram <exp></td><td>Sets PIMM1 SRAM margin, <exp> 0..255</td></tr>
<tr><td>network</td><td>Gets network communication interface status</td></tr>
<tr><td>network usb</td><td>Sets network communication interface to USB mode</td></tr>
<tr><td>network 422</td><td>Sets network communication interface to 422 mode</td></tr>
<tr><td>network autodetect|auto on|off</td><td>Turns network auto detection on or off</td></tr>
<tr><td>nmi</td><td>Sends NMI to nodes</td></tr>
<tr><td>nvram reset</td><td>Returns L1 settings to factory default</td></tr>
<tr><td>pbay</td><td>Gets the power bay status; use of this command requires the system management card</td></tr>
<tr><td>pbay dcport|dcprt</td><td>Gets power bay DCport status</td></tr>
<tr><td>pbay dcport|dcprt <str></td><td>Gets power bay DCport <<i>str</i>> status (<i>str</i>=1-8)</td></tr>
<tr><td>pbay dps</td><td>Gets power bay DPS status</td></tr>
<tr><td>pbay dps <str></td><td>Gets power bay DPS <<i>str</i>> states (<i>str</i>=1-6)</td></tr>
<tr><td>pbay env</td><td>Gets power bay env monitoring status</td></tr>
<tr><td>pbay env on|off</td><td>Enables or disables power bay env monitoring</td></tr>
</table>

Table 4-2 L1 Commands (Version 1.4.1)

Command	Description
`pbay fru|f`	Gets power bay and DPS FRU information; this command does not display DPS capacity information
`pbay fru <str>`	Gets power bay (*str*=0) and DPS (*str*=1-6) FRU information
`pbay init`	Manually initializes the power bay (ICMB interface)
`pbay reset|rst`	Resets power bay
`pbay test|tst`	Generates power bay self-test (future releases)
`pbay version|ver`	Gets power bay firmware version
`pci`	PCI status
`pci [u|d]`	Powers up or down all PCI buses and slots
`pci rst|reset`	Resets all PCI buses and slots
`pci` *<exp>* `[u|d]`	Powers bus *<exp>* up or down
`pci` *<exp>* `[reset]`	Resets bus *<exp>*
`pci` *<exp>* *<exp>* `[u|d]`	Powers up or down bus *<exp>* slot *<exp>*
`pci` *<exp>* *<exp>* `rst|reset`	Resets bus *<exp>* slot *<exp>*
`pimm`	Gets clock settings
`pimm auto`	Switches back to auto-selecting the clock source and mode based on the PIMM PSC code; this command is only needed if these settings were forced manually with the *pimm clksrc* or *pimm clkmode* settings. This is a hidden command.
`pimm clksrc ext|external`	Sets clock source external; the source is auto-set from PIMM PSC code. This is a hidden command.
`pimm clksrc int|internal`	Sets clock source internal; this is a hidden command. The source is auto-set from PIMM PSC code. This is a hidden command.
`pimm clkmode [async|sync]`	Sets asynchronous or synchronous clock mode; the source is auto-set from PIMM PSC code. This is a hidden command.
`port|prt`	Gets port status
`power|pwr`	Gets power status (verbose)
`power|pwr check`	Checks power status (terse)
`power|pwr [up|u] [down|d]`	Powers up or down

Table 4-2 L1 Commands (Version 1.4.1)

Command	Description
`power|pwr vrm`	Displays the status of the 48V fail warning and the -12V okay settings on the I, P and X bricks
`power|pwr up|u hold`	Powers up; holds reset
`reboot_l1`	Boots the default flash image
`reboot_l1 [a|b]`	Boots flash image A or B
`reboot_l1 current|old|new`	Boots the current, old, or new flash image
`reset|rst`	Resets the L1
`router|rtr`	Gets router information
`router|rtr meta`	Sets router type to meta
`router|rtr repeater|rep`	Sets router type to repeater
`router|rtr ordinary|ord`	Sets router type to regular
`router|rtr 6 port`	Disables two router ports
`router|rtr 8 port`	Enables all router ports
`router|rtr 8 port` *<str> <str> <str> <str>*	Enables all router ports
`router|rtr service <str>` *<str> <str> <str>*	Puts router in service mode for repair
`router|rtr service off`	Disables service mode after repairs are completed
`router|rtr spare`	Configures router for spare teardown
`scan`	
`scan reset`	Performs hard and soft JTAG reset
`scan reset hard`	Performs hard JTAG reset (via TRSTN)
`scan reset soft`	Performs soft JTAG reset (via TMS)
`scan reset both`	Performs hard and soft JTAG reset
`scan sel` *<exp> <exp>*	Selects SIC: *<addr> <CER>*
`scan sel` *<exp> <exp> <exp>*	Selects SIC: *<addr> <CER> <MR>*
`scan sel` *<exp> <exp> <exp> <exp>*	Selects SIC: *<addr> <CER> <MR> <IOR>*
`scan ids` *<exp>*	Reads and displays IDCODE registers
`scan set trst 0|1`	Directs control of JTAG TRST signal
`scan set tck 0|1`	Directs control of JTAG TCK signal
`scan set tms 0|1`	Directs control of JTAG TMS signal

Table 4-2 L1 Commands (Version 1.4.1)

Command	Description
scan set tdi 0|1	Directs control of JTAG TDI signal
scan set psi 0|1	Directs control of JTAG PSI signal
scan get	Displays state of JTAG TAP signals
scan debug *<exp>*	Sets scan debug message level
scan debug	Displays scan debug message level
scan info	Displays scan information
scan count	Counts length of IR and bypass registers
scan count *<exp>*	Counts length of IR and bypass registers
security	Tests the brick's security readiness
security verify 3400	Verifies correct serial number and router port settings for Origin/Onyx 3400 systems
security verify 3800	Verifies correct serial number and router port settings for Origin/Onyx 3800 systems
select|sel	Shows brick and subchannel for console I/O
select|sel *<exp>* *<exp>*	Sets rack and slot for console I/O, subchannel
select|sel console|con	Sets 'console' subchannel for console I/O
select|sel local	Sets local brick for console I/O, subchannel
select|sel ctc	Sets attached C brick for console I/O, subchannel
select|sel a|b|c|d	Sets selected CPU subchannel (0-3) for console I/O
select|sel *<exp>*	Sets selected CPU subchannel for console I/O
select|sel filter on|off	Enables or disables console output filter
serial	Shows secure system serial numbering information only
serial all	Shows system and brick part/serial numbers.
serial *<str>* *<str>* *<str>* *<str>*	Erases and reassigns system serial number by using temporary authenticator
serial clear	Temporary equivalent to serial *<str>* *<str>* *<str>* *<str>*
serial dimm	Shows DIMM part/serial numbers
serial enforce on|off	Turns SSN enforcement on or off
serial security on	Enables system serial number security
serial test	Verifies correct system serial number settings

Table 4-2 L1 Commands (Version 1.4.1)

<table>
<tr><th>Command</th><th>Description</th></tr>
<tr><td>serial verify</td><td>Tests the brick's readiness for secure serial numbering</td></tr>
<tr><td>softreset|softrst</td><td>Soft reset</td></tr>
<tr><td>set bedrock|bdrck ppp</td><td>Sets L1<->bedrock protocol to PPP mode</td></tr>
<tr><td>test</td><td></td></tr>
<tr><td>test|tst i2c</td><td>Runs I^{2}c test one pass</td></tr>
<tr><td>test|tst i2c <<i>exp</i>></td><td>Runs I^{2}c <<i>exp</i>> passes</td></tr>
<tr><td>test|tst ioexp set <<i>exp</i>> <<i>str</i>></td><td>At ioexp index <<i>exp</i>1>, sets to value <<i>exp</i>2></td></tr>
<tr><td>test|tst ioexp get <<i>exp</i>></td><td>Gets the value of ioexp at index <<i>exp</i>></td></tr>
<tr><td>test|tst ioexp get all</td><td>Gets the value of ioexp at index <<i>exp</i>></td></tr>
<tr><td>test|tst intr</td><td>Gets I^{2}C interrupt counts</td></tr>
<tr><td>test|tst exception|exc mem</td><td>Tests memory fault handling</td></tr>
<tr><td>test|tst exception|exc wdog</td><td>Tests software watchdog reset</td></tr>
<tr><td>test|tst exception|exc fatal</td><td>Tests fatal_error() call</td></tr>
<tr><td>uart</td><td>Shows uart status</td></tr>
<tr><td>usb</td><td>Shows USB status and displays any debugging statistics for L1 to L2 connections</td></tr>
<tr><td>verbose</td><td>Gets SMP prompt and character echo mode</td></tr>
<tr><td>verbose [0|1]</td><td>Enables/disables SMP prompt and character echo</td></tr>
<tr><td>version|ver</td><td>Returns the current firmware image version</td></tr>
</table>

4.1.1 Virtual Debug Switch Settings

The *debug* command gets the value of the virtual debug switches. The *debug <exp>* command sets the value for the debug switches. Table 4-3 lists the debug bit values and describes their function

Table 4-3 Virtual Debug Switch Settings

Diagnostic Testing Level (Hex Value)	Switch Number	Description
0		Normal Testing
1	1	No Testing
2	2	Heavy Testing
3	1 and 2	Manufacturing-level Testing
Diagnostic Output Level		
4	3	Verbose; information level set to verbose
Boot Stop Point		
0		Normal; normal setting, do not stop
08	4	Global POD
10	5	Local POD; boot stop requested at local POD
18	4 and 5	Memoryless POD; boot stop requested at no memory POD
Default Environment		
20	6	Ignore environment variable
Bypass IO7		
40	7	Bypasses first IO7
Bypass Global Master		
80	8	Gives up global master status
Override CPU Disabling		

Table 4-3 Virtual Debug Switch Settings

Diagnostic Testing Level (Hex Value)	Switch Number	Description
100	9	Overrides CPUs disabled with the environment variables in POD mode
Hardware Error State		
1000	14	Dumps hardware error state at system boot time
Ignore Autoboot		
2000	15	IO7 PROM ignores the autoboot environment variable

4.2 L2 Commands

Table 4-4 Connecting to the L2

USB Connection (Emulator)	Ethernet Connection	Description
1. `cd /stand/sysco/bin`	`cd /stand/sysco/bin`	Directory in which firmware commands reside
2. `./l2`	`./l2term`	Command to establish an L2 connection
3. `ctrl d`	`ctrl d`	Keystroke to get to the System Maintenance Menu from the L2 prompt

Note: To exit the software L2 type `quit`. To exit the hardware L2 type `ctrl-] + q`.

Table 4-5 L2 Commands (Version 1.4.2)

Command	Description
`autopower`	Displays current auto power-up status
`autopower|apwr on|off`	Enables or disables system auto power-up after L1 or L2 controller power up
`autopower|apwr abort`	Aborts auto power on
`config|cfg`	Shows configuration information
`config|cfg auto`	Enables automatic L1 config updates
`config|cfg devices|dev`	Shows configuration information w/devices
`config|cfg l2`	Shows L2 configuration information (normal)
`config|cfg man|manual`	Disables automatic L1 config updates
`config|cfg print|printall`	Used for debugging only
`config|cfg re|rescan`	Forces L1 config update
`config|cfg summary|s`	Shows configuration information (summary)
`config|cfg verbose|v`	Shows all information about connected L1's and L2's
`connect` *<str>*	Connects to L2 at host specified
`destination|dest`	Shows default destination(s)

Table 4-5 L2 Commands (Version 1.4.2)

<table>
<tr><th>Command</th><th>Description</th></tr>
<tr><td>rack|r <<i>rng</i>> [slot|s <<i>rng</i>>] destination|dest</td><td>Sets default destination to selected racks and slots</td></tr>
<tr><td>destination|dest reset|rst</td><td>Resets default to all rack and slots</td></tr>
<tr><td>help|hlp</td><td>Gets list of commands</td></tr>
<tr><td>help|hlp <str></td><td>Gets help on a single command</td></tr>
<tr><td>ip</td><td>Shows static IP address settings</td></tr>
<tr><td>ip <<i>str</i>> <<i>str</i>></td><td>Sets the L2's static IP config: <<i>addr</i>> <<i>netmask</i>></td></tr>
<tr><td>ip <<i>str</i>> <<i>str</i>> <<i>str</i>></td><td>Sets the L2's static IP config: <<i>addr</i>> <<i>netmask</i>> <<i>broadcast</i>></td></tr>
<tr><td>ip clear|reset</td><td>Clears static IP address settings</td></tr>
<tr><td>l1</td><td>Engages L1 command processor</td></tr>
<tr><td><rack>.<slot> l1 <<i>command</i>></td><td>Sends L1 command to selected L1</td></tr>
<tr><td>rack|r <<i>rng</i>> [slot|s <<i>rng</i>>] l1</td><td>Engages L1 command processor at rack and slot</td></tr>
<tr><td>:<<i>port</i>>:<l1> l1</td><td>Engages L1 command processor on port</td></tr>
<tr><td>rack|r <<i>rng</i>> [slot|s <<i>rng</i>>] l1 <<i>command</i>></td><td>Sends <i>command</i> to selected L1(s)</td></tr>
<tr><td>:<<i>port</i>>:<l1> l1 <<i>command</i>></td><td>Sends <i>command</i> to L1 on selected port.</td></tr>
<tr><td><ip>:<port>:<l1> l1 <<i>command</i>></td><td>Sends L1 command to L1 at IP:port:L1</td></tr>
<tr><td>l2</td><td>Engages L2 command processor</td></tr>
<tr><td>rack|r <<i>rng</i>> [slot|s <<i>rng</i>>] l2</td><td>Engages L2 command processor at specified rack</td></tr>
<tr><td>rack|r <<i>rng</i>> [slot|s <<i>rng</i>>] l2 <<i>command</i>></td><td>Sends L2 command to selected L2(s)</td></tr>
<tr><td><ip> l2 <<i>command</i>></td><td>Sends L2 command to L1 at IP</td></tr>
<tr><td>l2dbg</td><td>Shows current L2 debug states</td></tr>
<tr><td>l2dbg broadcast <<i>exp</i>></td><td>Sets broadcast debug level</td></tr>
<tr><td>l2dbg config <<i>exp</i>></td><td>Sets L2 configuration debug level</td></tr>
<tr><td>l2dbg discovery|disc <<i>exp</i>></td><td>Sets L2 discovery debug level</td></tr>
<tr><td>l2dbg e|estat</td><td>Reports Event processor stats</td></tr>
<tr><td>l2dbg e|estat r|reset</td><td>Reports Event processor stats and reset to zero</td></tr>
<tr><td>l2dbg malloc <<i>exp</i>></td><td>Sets irouter malloc/free debug level</td></tr>
<tr><td>l2dbg meminfo|mem</td><td>Displays dynamic memory usage</td></tr>
</table>

Table 4-5 L2 Commands (Version 1.4.2)

Command	Description
`l2dbg packets` *<exp>*	Sets irouter packet debug level
`l2dbg sbuf` *<exp> <exp>*	Sets L2<->L2 socket send/recv buffer size
`l2dbg smp` *<exp>*	Sets SMP input debug level
`l2dbg usb` *<exp>*	Sets L2 USB debug level (0=off,1=debug,2=verbose)
`l2dsp|l2display`	Shows enabled and disabled status of input from the L2 touchscreen display
`l2dsp|l2display debug` *<exp>*	Enables debugging messages for L2 display
`l2dsp enable/disable`	Enables or disables status of input from the L2 touchscreen display
`l2dsp|l2display enable|disable`	Enables or disables touch-screen input from the L2 display
`l2find`	Prints list of L2's on the same subnet as this one
`log`	Dumps log
`log reset`	Clears log
`log insert` *<command>*	Inserts entry into log
`loopback`	
`rack|r` *<rng>* `[slot|s` *<rng>*`] loopback` *<exp> <exp>*	Loops back data to L1
`multisys`	New name for "share net"; shows current settings for multiple system network sharing
`multisys on|off`	Enables or disables multiple systems on the same network; that is, it enables support for L2 controllers in multiple systems to stay separate from each other while connected to the same network
`nvram reset`	Sets all values in NVRAM to their defaults
`power|pwr`	Show power status of each selected brick
`power|pwr delay`	Shows delay between power commands
`power|pwr delay` *<exp>*	Sets delay between power commands in milliseconds
`power|pwr up|u|down|d`	Powers up or down each selected brick
`power|pwr summary|s`	Shows power status summary. Includes bricks powered up, down, and their margin states

Table 4-5 L2 Commands (Version 1.4.2)

Command	Description
rack|r <rng> [slot|s <rng>] power|pwr	Show power status of each selected brick
rack|r <rng> [slot|s <rng>] power|pwr up|u	Powers up each selected brick
rack|r <rng> [slot|s <rng>] power|pwr down|d	Powers down each selected brick
quit|<ctrl>] q	Exits the software L2. Type **ctrl-]+q** to exit the hardware L2.
rackid	Prints the rackid of the L2
rackid <decimal exp>	Sets the rack ID for the L2
reboot_l2	Reboots the L2 controller
reboot_l2 force	Reboots the L2 controller even if the firmware image is invalid
serial	Shows the L2 system serial number
serial <str>	Sets the L2 system serial number
select|sel	Shows brick, rack, slot and subchannel for console I/O
select|sel subchannel|sub|s console|con	Sets "console" subchannel for console I/O and current brick
select|sel subchannel|sub a|b|c|d	Sets selected CPU subchannel for console I/O and current brick
select|sel subchannel|sub|s <exp>	Sets selected CPU subchannel for console I/O, current brick
select|sel *<exp>* *<exp>*	Sets rack and slot for console I/O and subchannel
select|sel filter on|off	Enables or disables console output filter
select|sel *<R>*.*<S>*	Sets the system console input brick to rack R, slot S
select|sel partition|part|p <exp>	Sets the system console input brick to the console brick for the specified partition, and enables console output filtering; also sets the default command destination for the l2term/telnet session
select|sel syscon l2	L2 detects system console
select|sel syscon prom	PROM requests system console

Table 4-5 L2 Commands (Version 1.4.2)

Command	Description
select|sel subchannel	Sets the console subchannel for the l2term session (previous syntax was "select x", where x is the subchannel)
select|sel verbose	Reports the current system console information for each partition
shell|!	Escapes to L3/L2 OS
shell|! <command>	Sends command to the L3/L2 OS
smp	Shows the status of the SMP network connection and which sessions are connected
smp join	Allows an L2 command (*l2term/telnet*) session to join another session so that the user sees the keystrokes and output from the session
smp leave	Allows an L2 command (*l2term/telnet*) session to disconnect from a connected session and resume its own private session
smp reset	Resets the SMP network connection
syscon	Reports the current console brick for each partition
sysname	Shows the system name
sysname *<str>*	Sets the system name
tty	Shows terminal configuration
version|ver	Shows software version

4.3 PROM Monitor Commands

Table 4-6 PROM Monitor Commands (Version 6.54)

Command	Description
`auto`	Autoboots the machine
`arcshinv [-v] [-m] [-mvvv] [-g` *<path>*`]`	Displays the hardware inventory
`boot [-f` *file*`] [-n] [`*args*`]`	Loads and executes the specified file
`diag_enet [-N` *nasid*`] [-m` *moduleid*`] [-b` *bus*`] [-p` *pcidev*`] [-h] [-e] [-v] [-t` *testname*`] [-R | -r` *count*`]`	Runs the Ethernet power-on diagnostic; this is a hidden command `-N` *nasid* specifies the NASID number of the module to test (defaults to the console I brick); this option supersedes the NASID that is indicated by the module and slot values `-m` *moduleid* specifies the module number of the module to test (defaults to the console I brick) `-b` *bus* specifies the PCI bus to test. `-p` *pcidev* specifies the PCI device number of the target IOC3 `-h` runs the diagnostic in heavy mode `-e` runs the diagnostic in external loopback mode `-v` runs the diagnostic in verbose mode `-t` *testname* specifies the test name; valid names are ssram, txclk, phyreg, ioc3_loop, phy_loop, tw_loop, ext_loop, ext_loop_10, ext_loop_100, and xtalk (the default is to run all tests) `-R` *count* repeats the test count times; the test continues executing when it detects a failure `-r` *count* repeats the test count times; the test stops when it detects a failure

Table 4-6 PROM Monitor Commands (Version 6.54)

Command	Description

`diag_fibre`
`[-N` *nasid*`]`
`[-m` *moduleid*`]`
`[-b` *bus*`]`
`[-p` *pcidev*`] [-h] [-v]`
`[-t` *testname*`]`
`[-R | -r` *count*`]`
	Runs the fibre channel power-on diagnostic; this is a hidden command
	
	-N nasid specifies the NASID number of the module to test (defaults to the console I brick); this option supersedes the NASID that is indicated by the module and slot values
	
	`-m` *moduleid* specifies the module number of the module to test (defaults to the console I brick)
	
	`-b` *bus* specifies the PCI bus to test
	
	`-p` *pcidev* specifies the PCI device number of the target IOC3
	
	`-h` runs the diagnostic in heavy mode
	
	`-v` runs the diagnostic in verbose mode
	
	`-t` *testname* specifies the test name; valid names are *ram, dma, and controller* (the default is to run all tests)
	
	`-R` *count* repeats the test count times; the test continues executing when it detects a failure
	
	`-r` *count* repeats the test count times; the test stops when it detects a failure

`disable -m` *modid*`[-cpu [abcd]] [-mem [01234567]] [-s` *slotid* `-pci [01234567]]`	Disables hardware **Note:** If you disable CPU A, you must also disable CPU B; if you disable CPU B you do not need to disable CPU; this will be fixed in the 6.5.11 IRIX release
`enable -m` *modid* `[-cpu [abcd]] [-mem [01234567]] [-s` *slotid* `-pci [01234567]]`	Enables hardware
enableall [-y] [-list]	Enables all disabled components
`exit`	Returns to the PROM menu

Table 4-6 PROM Monitor Commands (Version 6.54)

Command	Description
`flash [-e] [-m` *modid*`] [-N` *nasid*`] [-f] [-F] [-y] [-v] [-e] [-l] [-C] FILE`	Flashes all appropriate PROMs with *file;* this is a hidden command
`help [`*command*`]`	Prints descriptions for one or all commands
`hinv [-v] [-m] [-mvvv] [-g` *<path>*`]`	Displays hardware inventory
`init`	Reinitializes the PROM log
`ls [`*device*`]`	Displays a directory listing for a device
`modnum`	Lists all the current modules in the system
opts	Displays the build options
`passwd`	Sets the PROM password.
`ping [-f] [-i` *sec*`] [-c` *num*`]` *ipaddr*	Pings server at target IP address
pod	Enters POD mode command interpreter
`printenv [env_var_list]`	Displays selected environment variables
`reset`	Resets the machine
`resetenv`	Resets the environment to standard values
`resetpw`	Resets the PROM password
serial	Displays the encrypted system serial number
`setenv` *env_var string*	Sets the value of an environment variable
setpart	Makes partitions
`single`	Boots IRIX OS to single-user mode
`unsetenv` *env_var*`:`	Removes a variable from the environment
`update`	Updates the stored hardware inventory
`version`	Displays PROM version

4.4 POD Commands

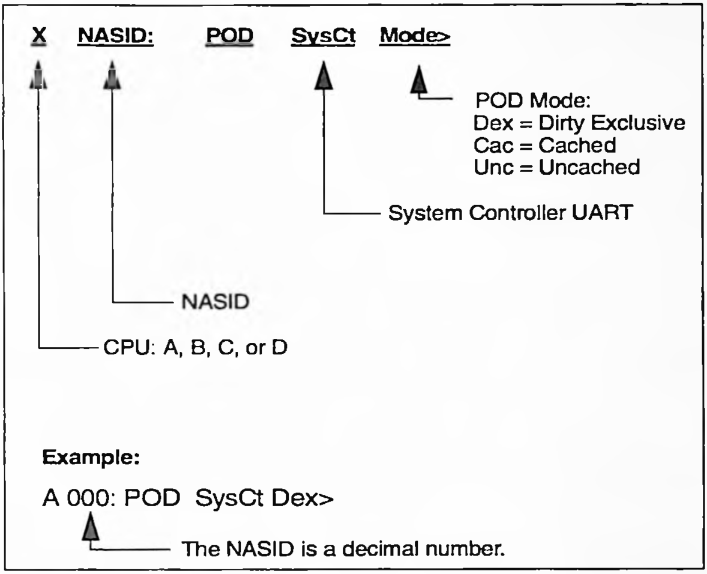


Figure 4-2 POD Prompt

Table 4-7 POD Mode Command Reference (Version 6.54)

Command/Syntax	Description
`?` [*cmdlname*]	Displays help information
`adline` *line*	Dumps the specified data cache line
`adtag` *line*	Dumps the specified data cache tag
`ailine` *line*	Dumps the specified instruction cache line
`aitag` *line*	Dumps the specified instruction cache tag
`altregs` *num*	Uses the alternate registers
`asline` *line*	Dumps the specified secondary cache line
`astag` *line*	Dumps the specified secondary cache tag
`btrace [epc sp]`	Stacks backtrace
`call` *addr* `[A0 [A1 [...]]]`	Calls a subroutine
`cfg [n:`*node*`]`	Displays the processor config register
`chklink`	Runs the local link connection test

Table 4-7 POD Mode Command Reference **(continued)** (Version 6.54)

Command/Syntax	Description
`clear`	Clears errors
`clearlog [n:`*node*`]`	Clears the log only; this command does not clear variables and aliases
`clearalllogs`	Clears all PROM logs in the system
`cpu [[n:`*node*`] a|b|]`	Switches to the specified CPU
`crb [n:`*node*`]`	Dumps I/O interface (II) CRBs
`crbx [n:`*node*`]`	Dumps the CRBs in a 137-column-wide format
`dbg [`*virt_val*`]`	Displays the virtual debug switch settings
`delay` *microsec*	Delays command execution
`dgbrdg [`*mn* `|` *h* `|` *m*`] [`*nnode*`] [`*sslot*`]`	Runs the Bridge test
`dgconf [`*mn* `|` *h* `|` *m*`] [`*nnode*`] [`*sslot*`]`	Runs the PCICONFIG test
`dgpci [`*mn* `|` *h* `|` *m*`] [`*nnode*`] [`*sslot*`] [`*ppcinum*`]`	Runs the PCI bus test
`dgsdma [`*mn* `|` *h* `|` *m* `|` *x*`] [`*nnode*`] [`*sslot*`]`	Runs the serial DMA test
`dgspio [`*mn* `|` *h* `|` *m* `|` *x*`] [`*nnode*`] [`*sslot*`] [`*ppcinum*`]`	Runs the serial programmed I/O test
`dgxbow [`*mn* `|` *h* `|` *m*`] [`*nnode*`]`	Runs the XBOW test.
`dips`	Displays the virtual debug switch settings
`dirinit` *start len*	Initializes a range of memory to the "Uknown" state
`dirstate [`*base* `[`*len* `[`*state* `]]`	Scans a range of addresses for the directory state that you specify
`dirtest` *start len*	Performs directory test and initialization
`dis` *addr* `[`*count*`]`	Disassembles the data stored in the specified memory locations
`disable n:`*node* `[`*slice/banks*`]`	Disables the specified CPU **Note:** If you disable CPU A, you must also disable CPU B. If you disable CPU B you do not need to disable CPU A; this will be fixed in the 6.5.11 release
`disc`	Discovers the NUMAlink interconnect
`dline` *line*	Dumps the specified data cache line
`dmc [n:`*node*`]`	Dumps memory configuration

Table 4-7 POD Mode Command Reference **(continued)** (Version 6.54)

Command/Syntax	Description
`dtag` *line*	Dumps the specified data cache tag
`dumpspool [n:`*node slice*`]`	Dumps the PI error spool
`ecc [on | off]`	Enables or disables ECC mode
`echo` *<string>*	Echoes a string on the console
`edump_bri [n:`*node*`]`	Dumps bridge error information to memory
`enable n:`*node* `[`*slice/banks*`]`	Enables the specified CPU
`error`	Displays errors
`error_dump`	Dumps the error information
`exec [`*segname* `[`*flag*`]]`	Loads and execute a segment
`flash` *node* `[...]`	Programs remote PROM
`fflash` *node* `[...]`	Prompts the user for processor specific values to flash
`flush`	Flushes and invalidates cache(s)
`for (`*cmd;expr;cmd*`)` *cmd*	Repeats the specified command(s) for the number of iterations determined by the specified expression
`fru [1 (local)|2 (all node)]`	Runs the FRU analyzer program
`go dex|unc|cac`	Sets memory mode
`help [`*cmdname*`]`	Displays help
`hubsde`	Runs the Hub send data error test
`if` *<exp> cmd*	Executes the specified command(s) depending on the value of a conditional expression
`iline` *line*	Dumps the specified instruction cache line
`im [`*byte*`]`	Sets the processor interrupt mask
`initalllogs`	Initializes all PROM logs for all modules in the system
`initlog [n:`*node*`]`	Initializes the PROM log
`inval [i] [d] [s]`	Invalidates cache(s)
`jump` *addr* `[A0 [A1]]`	Invalidates cache and continue execution at a specific address
`kdebug [`*stackaddr*`]`	Debugs the kernel
`kern_sym`	Uses the kernel symbol table
`la` *addr* `[`*count*`]`	Loads byte data and displays it as ASCII characters
`lb` *addr* `[`*count*`]`	Loads a byte

Table 4-7 POD Mode Command Reference **(continued)** (Version 6.54)

Command/Syntax	Description
`ld` *addr* [*count*]	Loads a double word
`lh` *addr* [*count*]	Loads a halfword
`log` [n:*node*] [*skip*[*count*]]	Displays entries from the PROM log for the specified node
`loop` *cmd*	Repeats the specified command(s) forever
`lw` *addr* [*count*]	Loads a word
`maxerr` *count*	Tests the error limit
`mbist` [*addr*]	Memory built-in self-test
`memcmp` *dst src len*	Compares data (by bytes) in one memory location to data in another memory location
`memcpy` *dst src len*	Copies memory (by bytes) from one memory location to another memory location
`meminit` *start len*	Clears memory
`memset` *dst byte len*	Fills memory with a byte pattern
`memsum` *src len*	Adds data (by bytes)
`memtest` *start len*	Performs a memory sanity test
`modnic`	Displays the NIC value of the current module
`module` [*num*]	Displays or change a module number
`nic` [n:*node*]	Displays the Hub NIC
`nm` *addr*	Looks up a PROM address
`nmi` n:*node* [a|b|c|d]	Sends NMI to node
`node` [[*vec*] *id*]	Gets or sets node ID
`pa` *addr* [*bitno*]	Prints an address
`partition` [*num*]	Gets/sets a hardware partition number
`pb` *expr*	Prints the value of an expression in binary
`pcfg` [n:*node*]	Dumps the pcfg structure. Not supported in dex mode
`pd` *<exp>*	Prints the value of an expression in decimal
`pf` [*regno*]	Prints the contents of a processor's COP1 floating-point register
`po` *<exp>*	Prints the value of an expression in octal
`pod`	Invokes POD mode command interpreter

Table 4-7 POD Mode Command Reference **(continued)** (Version 6.54)

Command/Syntax	Description
pr [*gprname* [*val*]	Prints the contents of a register
printenv [n:*node*] [*key*]	Prints the value of the specified environment variable(s)
px *<expr>*	Prints the value of an expression in hexadecimal
qual [on | off]	Enables or disables quality mode
reconf	Initializes the memory/directory section (MD) of the Hub
repeat *count cmd*	Repeats count
reset [*all*]	Resets the system
rnic [*vec*]	Reads the router NIC
route [*vec node*]	Sets up a route
rstat *vec*	Dumps and clears the router status
rtab [*vec*]	Dumps the route table
rtrsde	Runs the router send data error test
santest *addr*	Runs the memory sanity test
sb *addr* [*val*[*count*]]	Stores a byte
scandir *addr* [*len*]	Scans directory states
sd *addr* [*val* [*count*]]	Stores a double word
sdv *addr count*	Stores a double word and verifies it
segs [*flag*]	Lists the segments
setenv [n:*nasid*] *var*[*<string>*]	Sets a environment variable
sf *regno val*	Stores the contents of a processor's COP1 floating-point register
sh *addr* [*val* [*count*]]	Stores a halfword
slave	Enters slave mode
sleep *sec*	Sleeps for a specified number of seconds
sline *line*	Dumps the specified secondary cache line
softreset n:*node*	Performs a soft reset on a node
sr *reg val*	Stores the contents of a register
stag *line*	Dumps the specified secondary cache tag
sstag *line taglo taghi* [*way*]	Stores a cache tag
sscache *line taglo taghi*	Stores an scache double word
sw ADDR [*val*[*count*]]	Stores a word
talk [n:*node* a|b]	Uses the net UART
tdisable n:*node* [*slice*]	Temporarily disables the specified CPU
time *cmd*	Performs benchmark timing

Table 4-7 POD Mode Command Reference **(continued)** (Version 6.54)

Command/Syntax	Description
`tlbc [`*index*`]`	Clears the TLB
`tlbr [`*index*`]`	Reads the TLB
`unsetenv [n:`*node*`]` *"var"*	Removes an environment variable
`verbose [0	1]`	Specifies whether or not the power-on diagnostics run in verbose mode
`version`	Displays the PROM version
`vr` *vec vaddr*	Performs a vector read operation
`vw` *vec vaddr val*	Performs a vector write operation
`while` *<exp> cmd*	Repeats the specified command(s) while the specified expression evaluates to true
`why`	Displays the current exception and NMI status

4.5 IRIX Commands

Table 4-8 lists IRIX commands commonly used by field personnel. For a complete list and description of IRIX commands, refer to the IRIX man pages or the IRIX user commands in the Technical Publications library at the following URL:

http://techpubs.engr.sgi.com/library/tpl/cgi-bin/browse.cgi?db=man&coll=0650&pth=/cat1

Table 4-8 Useful IRIX Commands

Command	Description
`sesmgr`	Fibre channel drive enclosure status/configuration command line interface
`getversion`	Reports the version level of *sesmgr*
`getstatus`	Reports status of supported fibre channel drive enclosures
`poll`	Requests that the daemon rescan the SAN for its configuration and also update the status of all known devices; typically this should need to be done only if the SAN configuration is manually changed outside of the *sesmgr* command set
`topology`	Prints the daemon's understanding SAN topology of the
`remove`	Allows for physical removal of a drive

Table 4-8 Useful IRIX Commands

Command	Description
insert	Allows for inserting a drive into a specified enclosure
ledon | ledoff	Turns "ON" or "OFF the indicator "LED" associated with the specified disk device
bypass	Performs an enclosure bypass to remove the specified disk on the specified port from the fibre channel loop
unbypass	Performs an enclosure unbypass to add the specified disk on the specified port onto the fibre channel loop
alarmon	Sounds the alarm on the specified enclosure
alarmoff	Turns off the alarm on the specified enclosure provided that the enclosure is clear of any other hardware conditions which also result in an alarm sounding
alarmmute	Mutes the enclosure's alarm; generally this produces a periodic reminder tone instead of totally silencing it
swap	Provides a method of adding, deleting, and monitoring the system swap areas used by the memory manager
osview	Monitors various portions of the activity of the operating system and displays them using the full screen capabilities of the current terminal
gmemusage	A graphical memory usage viewer
hinv	Displays the contents of the system hardware inventory table; this table is created each time the system is booted and contains entries describing various pieces of hardware in the system
linkstat	A NUMAlink monitoring tool that reports router link performance and error rates; it also reports error rates on the Hub chip NUMAlink network interface (NI) and I/O interface (II) links
chkconfig	Prints the state (on or off) of every configuration flag found in the directory */var/config*; flags normally are shown sorted by name; with the -s option they are shown sorted by state
mkpart	The partition administration tool for the Origin 3000 series of servers
topology	Machine topology information that is extracted from the hardware graph

Chapter 5

System Procedures

Subsection	Page Number
Establishing System Controller Connections	page 5-2
SGI 3200 Systems	page 5-2
SGI 3400 and 3800 Systems	page 5-3
Flashing System Controller Firmware	page 5-5
Obtain Updated L3 Software Revisions	page 5-5
Flash L1 Firmware	page 5-6
Flash L2 Firmware	page 5-6
Flashing IP35 Firmware	page 5-7
Flashing IP35 Firmware from a CD-ROM	page 5-7
Flashing IP35 Firmware from the PROM Monitor	page 5-7

5.1 Establishing System Controller Connections

The following subsections provide procedures for establishing connections to the L1 and L2 controllers from a Linux platform.

5.1.1 SGI 3200 Systems

5.1.1.1 Serial Connection to the L1 Controller

1. Ensure that network settings are set to the following values:
   - vt100
   - Baud 38400
   - No parity
   - 8 data bits
   - 1 stop bit
   - Hardware flow control on (rts/cts)
2. Connect a null modem serial cable from the L3 serial connector to the console connector on the rear of the C brick.
3. Type `cu -l ttyS0 -s 38400` at the Linux prompt. The L1 responds with "Connected."

```
$ cu -l ttyS0 -s 38400
Connected.
```

4. Press the Enter key to get to the L1 prompt.

```
001c04-L1>
```

5.1.1.2 USB Connection to the L2 Emulator

1. Connect a USB cable from the L3 controller to the L1 port (USB port) on the rear of the C brick.
2. Ensure that you are in the `/stand/sysco/bin` directory.
3. Type `./l2` at the Linux prompt.

```
$ ./l2
```

5.1.2 SGI 3400 and 3800 Systems

5.1.2.1 Serial Connection to the L2 Controller

1. Ensure that network settings are set to the following values:
   - vt100
   - Baud 38400
   - No parity
   - 8 data bits
   - 1 stop bit
   - Hardware flow control on (rts/cts)
2. Connect a null modem serial cable from the L3 serial connector to the console connector on the rear of the C brick.
3. Type **cu -l ttyS0 -s 38400** at the Linux prompt. The L1 responds with "Connected."

   ```
   $ cu -l ttyS0 -s 38400
   Connected.
   ```

4. Press the Enter key to get to the L2 prompt.

   ```
   001-L2>
   ```

5. Type **cfg** at the L2 prompt to display a list of all attached bricks.

   ```
   001-L2>cfg
   L2 01.1.1.77: - 001
   L1 10.1.1.77:0:0 - 001r19
   L1 10.1.1.77:1:0 - 001c07
   L1 10.1.1.77:1:1 - 001i21
   L1 10.1.1.77:2:0 - 001c10
   ```

5.1.2.2 Ethernet Connection to the L2 Controller

1. Connect an Ethernet cable from the L3 Ethernet port to the ENET connector on the back of the L2 controller:
   - Use a straight-through Ethernet cable for systems in which an Ethernet hub is used.
   - Use a crossover Ethernet cable if no Ethernet hub is used.
2. Reboot or restart the L3 network.

a. Type `/etc/rc.d/init.d/network stop` at the Linux prompt.

b. Type `/etc/rc.d/init.d/network start` at the Linux prompt.

3. Ensure that you are in the `/stand/sysco/bin` directory.
4. Type `./l2term` at the Linux prompt to establish a connection.

5.2 Flashing System Controller Firmware

The *flashsc* command and the system controller binary images reside on the L3 at the following location:

```
/usr/cpu/firmware/sysco
```

The following *flashsc* and *l2term* options are only required when multiple systems are connected to the same L3 controller.

Table 5-1 *flashsc* and *l2term* Command Options

Option	Description
–l2	<IP address/host>
–ssn	<system serial number>
–sysname	<system name> Assigned by using the L2 *sysname* command

5.2.1 Obtain Updated L3 Software Revisions

L3 software revisions are located at the following URL:

http://ist.csd.sgi.com/

Note: For your laptop or Linux platform to function as an L3 controller, the following Red Hat packages must be installed:

- pdksh
- uucp
- inetd
- telnet-server

1. Create a directory for the L3 software files:

   `% mkdir l3_date`

2. Open a web browser and access the following URL:

   `http://ist.csd.sgi.com`

3. Click on *Internal Support Tools CD*; then, select the CD release name that you want to download.
4. Click on `Linux` to access the Linux operation system releases.
5. Follow the installation steps for the release you want to download.

6. Reboot the L3.

5.2.2 Flash L1 Firmware

In the L2term window

1. Type the **flash default current* command at the L2 prompt. This command sets the image of all L1 controllers to the default image.

```
L2> * flash default current
```

In an L3 Linux window

2. Change to the `/usr/cpu/firmware/sysco` directory:

```
$ cd /usr/cpu/firmware/sysco
```

3. Type *./flashsc -p ./l1.bin all* command at the L3 prompt to flash all L1 controllers in parallel, type */flashsc ./l1.bin local* to flash a single brick, or type *flashsc --dev/dev/ttyS0 l1.bin* <r.s> to flash L1 firmware in a 3200 system using a serial connection.

```
$ ./flashsc -p ./l1.bin all
or
$ ./flashsc ./l1.bin <rack.slot>

$ ./flashsc --dev/dev/ttyS0 l1.bin <rack.slot>
```

In the L2term window

4. Type **reboot_l1 new* command at the L2 prompt. By using the *new* option in the command, you can test the new firmware.

```
L2> * reboot_l1 new
```

5. Type the *flash default reset* command at the L2 prompt to reset to the default image. This command picks (by date) the latest firmware image.

```
L2> * flash default reset
```

5.2.3 Flash L2 Firmware

1. Establish a connection between the L3 controller (Linux platform) and the system. Refer to 5.1.

In an L3 Linux window:

2. Change to the `/usr/cpu/firmware/sysco` directory:

```
$ cd /usr/cpu/firmware/sysco
```

3. Type *./flashsc -p ./l2.bin all* command at the L3 prompt to flash all L2 controllers in parallel or type *./flashsc ./l2.bin local* to flash a single brick.

```
$ ./flashsc -p ./l2.bin all
or
$ ./flashsc ./l2.bin local
```

In the L2term window:

4. Type * *reboot_l2* command at the L2 prompt.

```
L2> * reboot_l2
```

5.3 Flashing IP35 Firmware

Use the *flash* command at the IRIX prompt to flash firmware from the IRIX operating system.

```
# flash
```

5.3.1 Flashing IP35 Firmware from a CD-ROM

Use the *flashcpu cdrom (0, 6, 7) pathname* command to flash firmware from the IRIX CD-ROM.

```
$ flashcpu cdrom (0, 6, 7)/pathname
```

5.3.2 Flashing IP35 Firmware from the PROM Monitor

Use the *flashcpu* command to flash IP35 firmware from the PROM monitor.

```
> flashcpu
```

Chapter 6

Troubleshooting

Subsection	Page Number
Interpreting the LEDs Command Values	page 6-2
Cable Pinouts	page 6-8
48-V Power Connector	page 6-8
DNET Cable Connector and Pinouts	page 6-10
NASID to CPU Decoding	page 6-12
Troubleshooting a System Hang	page 6-13

6.1 Interpreting the LEDs Command Values

The L1 *led* command displays the status of the C-brick LED values. If an LED value is between 0x00 and 0x70, the message indicates a power-on LED or PLED. If the LED value is between 0x81 and 0xb5, the value indicates a FLED or failure LED. Useful POD values are listed in Table 6-1. LED values are listed in Table 6-2.

LED values only apply to PROM boot; once in IRIX, the LED value is only a meter for processor usage.

Table 6-1 Useful POD Codes

Code/Value	Description
00 or 45	In slave loop; 0x00/0x45=okay; solid 0x00/0x45=possible hang
80 or BC	In POD mode; the flashing of 0x80 and 0xbc indicates PROM is waiting for input on the UART channel
9a	CPUs disabled
98	No memory
55	Global master in PROM

Table 6-2 LED Command Values and Descriptions

Hex Value	Description
0x00	In slave loop; 0x00/0x45=okay; solid 0x00=possible hang
0x01	Initializes the processor, FPRS, and COP0 registers
0x02	Tests processor COP1 registers
0x03	Switches to mapped mode
0x04	Tests processor primary instruction cache
0x05	Tests processor primary data cache
0x06	Tests secondary cache
0x07	Flushes all caches
0x08	Not used
0X09	Not used

Table 6-2 LED Command Values and Descriptions

Hex Value	Description
0X0A	Invalidates processor primary instruction cache
0X0B	Invalidates processor primary data cache
0X0C	Invalidatse secondary cache
0X0D	Successfully jumped to the main () function
0X0E	About to increase PROM access speed
0X0F	Successfully increased PROM access speed
0X10	Not used
0X11	Not used
0X12	Not used
0X13	Not used
0X14	Not used
0X15	Not used
0X16	Not used
0X17	Not used
0X18	UART putc timed out
0X19	Not used
0X1A	Not used
0X1B	Not used
0X1C	Not used
0X1D	About to initialize selected UART.
0X1E	Done initializing selected UART.
0X1F	Not used
0X20	Not used
0X21	About to enter POD mode, C portion
0X22	About to enter POD prompt loop
0X23	About to enter POD mode, assembler portion
0X24	Performing local arbitration (CPU A/B)
0X25	Not used

Table 6-2 LED Command Values and Descriptions

Hex Value	Description
0X26	Not used
0X27	Not used
0X28	About to perform first local barrier
0X29	Not used
0X2A	About to configure Dex mode stack and data
0X2B	Reached main ()
0X2C	Not used
0X2D	Not used
0X2E	Not used
0X2F	Not used
0X30	Not used
0X31	In entry because of NMI
0X32	Not used
0X33	Not used
0X34	Not used
0X35	About to initialize Hub real-time counter
0X36	Done initializing Hub real-time counter
0X37	Not used
0X38	First local barrier succeeded
0X39	Not used
0X3A	Not used
0X3B	Not used
0X3C	About to jump to UALIAS space
0X3D	Jumped to UALIAS space
0X3E	About to jump to cached space
0X3F	Jumped to cached space
0X41	Done testing stack area of memory
0X42	Not used
0X43	Not used
0X44	Not used

Table 6-2 LED Command Values and Descriptions

Hex Value	Description
0X45	Slave loop; 0x00/0x45=okay; solid 0x45=possible hang
0X46	Received launch interrupt
0X47	Calling launched function
0X48	Launched function returned
0X49	Not used
0X4A	About to initialize Hub MD and SIMM controls
0X4B	About to probe and configure memory size
0X4C	Not used
0X4D	Not used
0X4E	Not used
0X4F	About to discover Hub I/O
0X50	About to update KLCONFIG for hubs and hubless routers
0X51	About to writer router cfg info into KLCONFIG
0X52	About to initialize I/O section of Hub
0X53	About to probe I/O section for console
0X54	Console probing completed
0X55	Not used
0X56	Done initializing I/O section of Hub
0X57	Reset error state saved
0X58	Hub error registers cleared
0X59	Hub error checking enabled
0X5A	Done discovering Hub I/O
0X5B	About to initialize NMI handler area
0X5C	About to test Hub interrupts
0X5D	About to perform early reset of Hub I/O section
0X5E	CPU came out of reset

Table 6-2 LED Command Values and Descriptions

Hex Value	Description
0X5F	Going to reset I^2C
0X60	Finished resetting I^2C
0X70	Running bist on bank 0
0X71	Running bist on bank 1
0X72	Running bist on bank 2
0X73	Running bist on bank 3
0X74	Running bist on bank 4
0X75	Running bist on bank 5
0X76	Running bist on bank 6
0X77	Running bist on bank 7
0X81	CP1 failed
0X82	Restart master unable to load IO7 PROM
0X83	icache failed
0X84	icache failed
0X85	scache failed
0X86	CPU disabled by another node
0X87	Real-time counter broken
0X88	Not used
0X89	Not used
0X8A	Not used
0X8B	Not used
0X8C	Not used
0X8D	Not used
0X8E	Not used
0X8F	OS requested LEDs
0X90	Not used
0X91	Hub local failed
0X92	Not used
0X93	Some node not premium
0X94	Not used
0X95	Not used

Table 6-2 LED Command Values and Descriptions

Hex Value	Description
0X96	Not used
0X97	Main () returned
0X98	No local memory
0X99	I^2C can't happen error
0X9A	CPU is disabled by environment variable
0X9B	Memory download failed
0X9C	Can not set core debug register
0X9D	Not used
0X9E	Hub klconfig failed
0X9F	Router klconfig failed
0XA0	Hub I/O init failed
0XA1	Node klconfig failed
0XA2	Not used
0XA3	Not used
0XA4	Hub chip failed 1/abist
0XA5	Not used
0XA6	Waiting for reset to go
0XA7	LLP failed after reset
0XA8	LLP never up after reset
0XA9	Not used
0XAA	Not used
0XAB	Network discovery failed
0XAC	NASID calculation failed
0XAD	Route calculation failed
0XAE	Route distribution failed
0XAF	NASID distribution failed
OXBO	Master assigned no NASID
OXB1	NASID arbitration failed
0XB2	Not used
0XB3	Not used
0XB4	Error copying mode bits

Table 6-2 LED Command Values and Descriptions

Hex Value	Description
0XB5	Not used
0XFE	TBD
0XFF	Console poll found data for reading

6.2 Cable Pinouts

6.2.1 48-V Power Connector

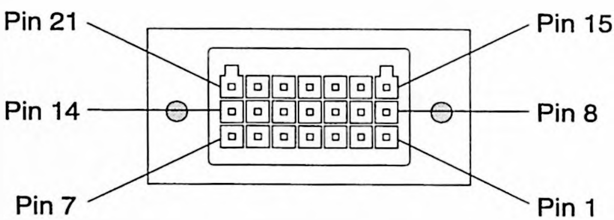


Figure 6-1 48-V Power Connector Pinouts (from Rear of Power Bay)

Table 6-3 DC Power Port Connector Pinouts

Pin Number	Pin Assignment
1	48-V
2	Tx/Rx(-)
3	Tx/Rx (+)
4	EMC Ground
5	ID (-)
6	ID (+)
7	48-V Return
8	48-V
9	48-V
10	Reserved
11	PB Port
12	Reserved
13	48-V Return

Table 6-3 DC Power Port Connector Pinouts **(continued)**

Pin Number	Pin Assignment
14	48-V Return
15	48-V
16	48-V
17	12-VSB return (L1)
18	48-V_EN
19	12-VSB (L1)
20	48-V Return
21	48-V Return

6.2.2 DNET Cable Connector and Pinouts

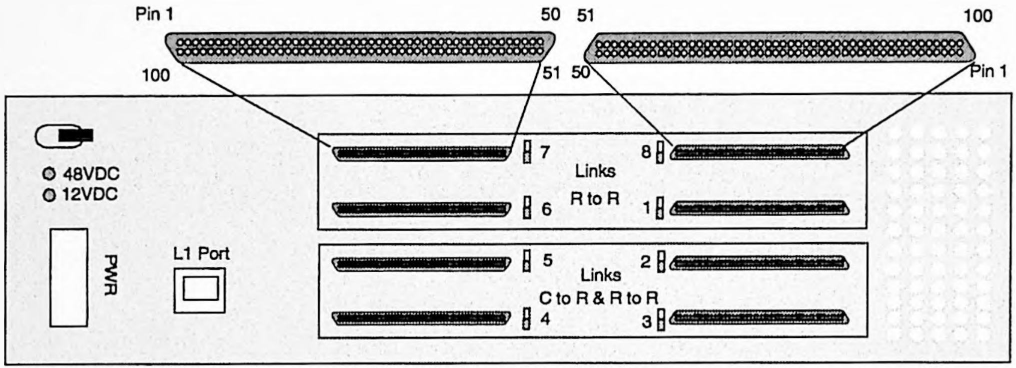
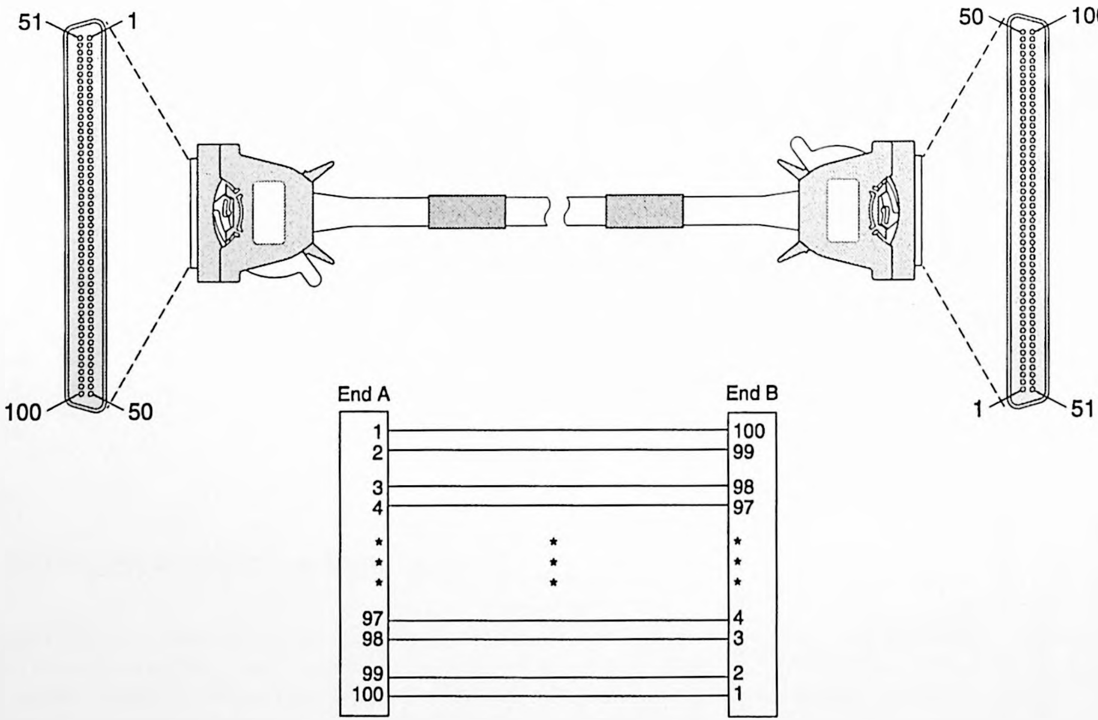


Figure 6-2 DNET Receptacle Pinouts (R Brick)



Wiring diagram

Figure 6-3 DNET Cable Connector Pinouts

6.3 NASID to CPU Decoding

6.3.1 3200 Systems

Table 6-4 3200 System NASID to CPU Decoding

NASID	CPU	U/Bay
1	4—7	C7
0	0—3	C4

001

6.3.2 3400 Systems

Table 6-5 3400 System NASID to CPU Decoding

NASID	CPU	U/Bay
7	28—31	C35
6	24—27	C32
5	20—23	C29
4	16—19	C24
3	12—15	C21
2	8—11	C16
1	4—7	C13
0	0—3	C10

001

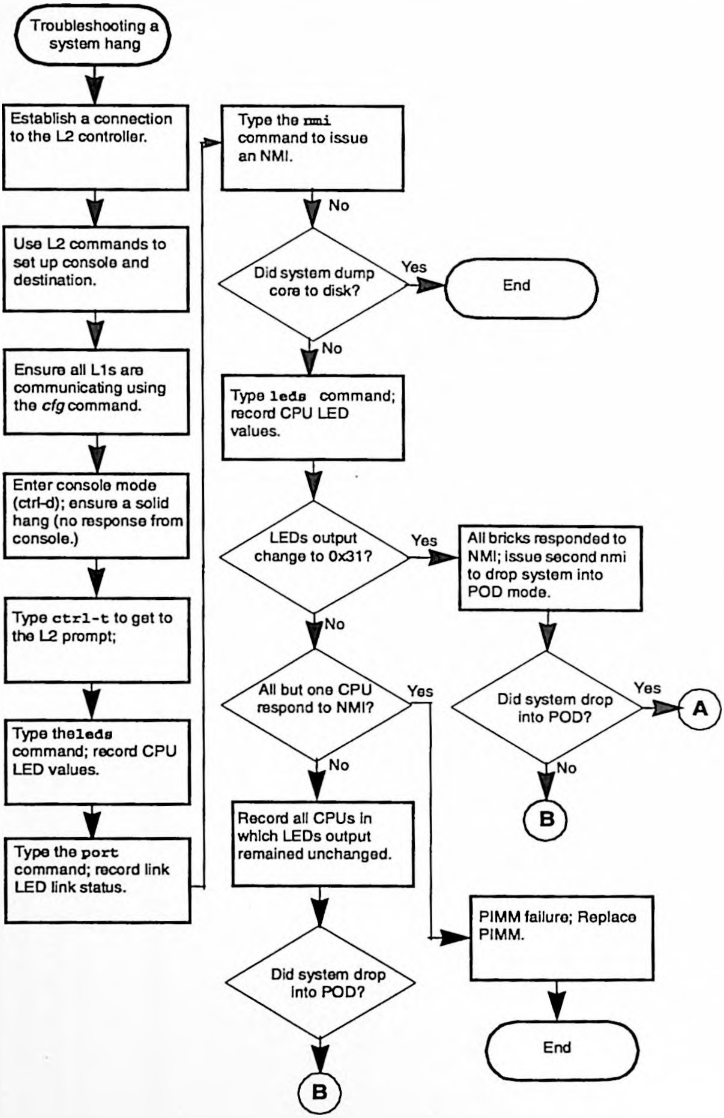
6.3.3 3800 Systems

Table 6-6 3800 System NASID to CPU Decoding

NASID	CPU	U/Bay	NASID	CPU	U/Bay	NASID	CPU	U/Bay	NASID	CPU	U/Bay
7	28—31	C35	15	60—63	C35	23	92—95	C35	24	124—127	C35
6	24—27	C32	14	56—59	C32	22	88—91	C32	23	120—123	C32
5	20—23	C29	13	52—55	C29	21	84—87	C29	22	116—119	C29
4	16—19	C24	12	48—51	C24	20	80—83	C24	21	112—115	C24
3	12—15	C21	11	44—47	C21	19	76—79	C21	20	108—111	C21
2	8—11	C16	10	40—43	C16	18	72—75	C16	19	104—107	C16
1	4—7	C13	9	36—39	C13	17	68—71	C13	18	100—103	C13
0	0—3	C10	8	32—35	C10	16	64—67	C10	17	96—99	C10
	001			002			003			004	

6.4 Troubleshooting a System Hang

Figure 6-4 is a flowchart for troubleshooting a system hang. This process helps you collect data in the Hardware Error State that may help you or an analyst isolate a system problem. Refer to PRB 200235 for detailed information on troubleshooting an Origin 3000 system hang. Refer to PRB number 200235 for detailed instructions on troubleshooting a system hang.



Refer to PRB Number 200235 for a detailed procedure on troubleshooting a hang.

Figure 6-4 Flowchart for Troubleshooting a System Hang

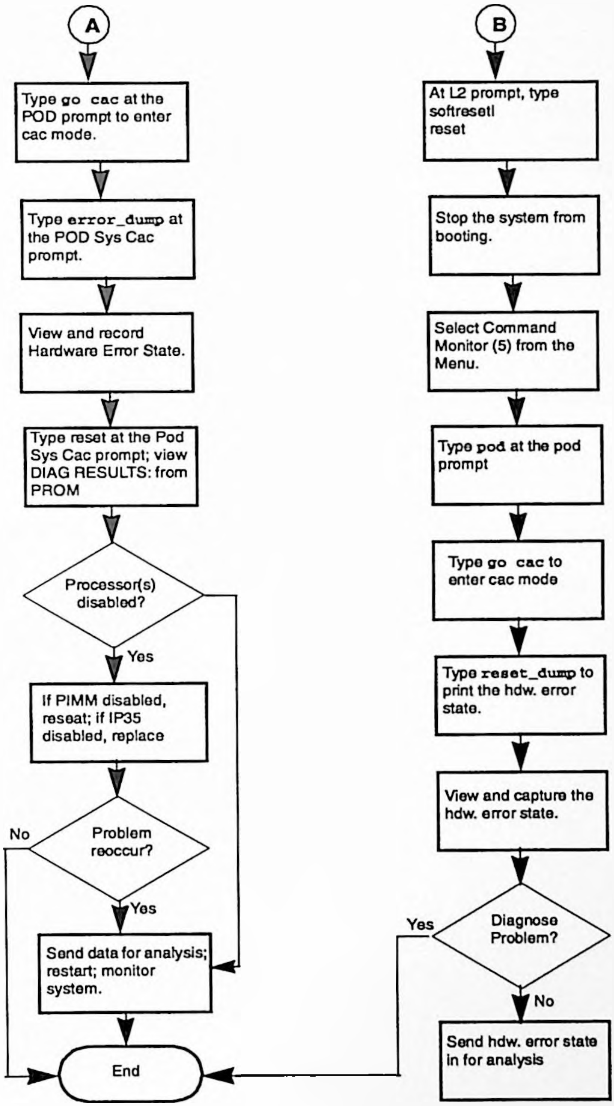


Figure 6-5 Troubleshooting Flowchart Continued (refer to Figure 6-4)

Chapter 7

FRU and Parts List

The following pages provide a list of the part numbers and names for all field replaceable units (FRUs) and spare parts that you can order. Refer to the following URL to obtain an up-to-date FRU list:

http://www-logistics.csd/cf/prod/spares/o3kspare.htm

An online illustrated parts catalog is available at the following URL:

http://servinfo.csd.sgi.com/sgi3000/sysmaint.htm

For instructions on how to replace these components, refer to the *SGI Origin 3000 Series Hardware Replacement Procedures*, document number 108-xxxx-001.

Subsection	Page Number
C Brick (013-3128-003)	page 7-2
I Brick (013-3035-002)	page 7-3
R Brick (013-3037-002)	page 7-5
P Brick (013-3034-002)	page 7-6
X Brick (013-3036-003)	page 7-8
D Brick	page 7-9
G Brick	page 7-10
Cables	page 7-11
Power Components	page 7-11
Misc. Support Equipment	page 7-12
Brick Hardware	page 7-12
Rack and Trim, 39U	page 7-12
Rack and Trim, 17U	page 7-14
PCI Options	page 7-15
XIO Options	page 7-15
Tools List	page 7-16

7.1 C Brick (013-3128-003)

Table 7-1 C-brick FRUs

Pilot Part Number	Description	Unit Qty	FE/FR [a]
013-2800-001	FAN ASSY HOT SWAP 3-4U	3	FE
013-3051-001	ENCLOSURE FRAME FRONT ¾U (FAN BOX)	1	FE
013-3052-001	CABLE DOCK SHIELDED	2	FE
013-3070-001	ENCLOSURE ASSY C-BRICK	1	FE
013-3071-001	LOGIC CARRIER ASSY S/M TREX	1	FE
015-0346-001	HARNESS ASSY ON/OFF CBRICK W/SWITCH	1	FE
015-0356-001	HARNESS ASSY EXT TO FANS SC P1	1	FE
015-0359-001	HARNESS ASSY C TREX INT MB TO BULK	1	FE
030-1018-002	PCA 256MB DIMM 128MB DDR	8	FR
030-1042-003	PCA 512MB DIMM 128MB DDR (PREMIUM)	8	FR
030-1044-001	PCA 512MB DIMM 128MB DDR	8	FR
030-1060-003	PCA 1GB DIMM 256MB DDR	8	FR
030-1439-002	PCA L1 SYSTEM CONTROL DISPLAY P1	1	FR
030-1604-004	PCA IP35 4CPU TREX (P1)	1	FR
030-1616-002	PCA IP35 PIMM R12KS 8MB 400MHZ	2	FR
040-3201-001	SHELF MODULE	1	FE
042-0708-003	GUIDE PIN DNET BR-NI .140DIA 2-FLAT	4	FE
042-0760-001	WASHER SHLDR INSUL .148ID .250OD	4	FE
042-0806-001	RACK EAR RIGHT DIE CAST	1	FE
042-0807-001	RACK EAR LEFT DIE CAST	1	FE
050-0616-001	HANDLE FAN PULL ¾U	3	FE
050-0619-001	PANEL ASSY FRONT	1	FE
050-0622-001	COVER CONTROL 3U	1	FE
050-0624-001	END CAP RIGHT	1	FE
050-0625-001	END CAP LEFT	1	FE
050-0654-001	CLIP PCB MOUNTING .105 THICK BOARD	3	FE
050-0654-001	CLIP PCB MOUNTING .105 BRD 10MM HT	1	FE
050-0683-001	LEVER SERVICE LEFT	1	FE
050-0684-001	LEVER SERVICE RIGHT	1	FE

Table 7-1 C-brick FRUs **(continued)**

Pilot Part Number	Description	Unit Qty	FE/FR [a]
050-0707-001	GUIDE PIN AND SPACER PIMM SNAP-IN	4	FE
050-0718-001	SUPPORT MEG-ARRAY CONN C-BRICK	1	FE
060-0104-001	POWER SUPPLY D2D 48V INP 3.3V 30A	2	FR
060-0107-001	POWER SUPPLY D2D 48V INP 2.5V 30A	3	FR
060-0108-001	POWER SUPPLY D2D 48V INP 1.8V 30A	2	FR

[a]FE=field expense; indicates that the item may be scrapped. FR=field repairable; indicates that the component must be sent back for repair.

7.2 I Brick (013-3035-002)

Table 7-2 I-brick FRUs

Part Number	Description	Unit Qty	FE/FR [a]
013-2614-002	CARRIER ASSEMBLY PCI	5	FE
013-2800-001	FAN ASSY HOT SWAP 3-4U	3	FE
013-2846-001	COVER ASSY DISK IO GEN	1	FE
013-3026-001	CARRIER ASSY PWR IO GEN	1	FE
013-3051-001	ENCLOSURE FRAME FRONT ¾U	1	FE
013-3052-001	CABLE DOCK SHIELDED	2	FE
013-3059-002	ENCLOSURE ASSY DISK IO	1	FE
013-3064-001	COVER ASSY REAR I-BRICK	1	FE
013-3095-001	CARRIER ASSY REAR IO GEN	1	FE
015-0323-001	HARNESS ASSY L1 INLET TO PWR BRD	1	FE
015-0342-001	HARNESS ASSY PWR INLET TO PWR BOARD	1	FE
015-0356-001	HARNESS ASSY EXT TO FANS SC P1	1	FE
015-0358-001	HARNESS ASSY PIX INT L1 TO FANS	1	FE
015-0364-001	HARNESS ASSY DISK POWER IO GEN T310	1	FE
018-0840-001	CABLE ASSY REM MED JUMPER	1	FE
018-0841-001	CABLE ASSY IEEE1394 6-6 .7M	1	FE
018-0896-001	CABLE ASSY SERIAL/ST	1	FE
018-0923-001	CABLE ASSY SDC 9PDSUB 600MM	1	FE
030-1439-002	PCA L1 SYSTEM CONTROL DISPLAY P1	1	FR

Table 7-2 I-brick FRUs **(continued)**

Part Number	Description	Unit Qty	FE/FR[a]
030-1557-007	PCA IO GENERAL P1	1	FR
030-1571-002	PCA POWER I/X BRICK	1	FR
030-1632-001	PCA POWER INPUT 48V 30/2A	1	FR
030-1637-002	PCA FC BULKHEAD	1	FR
030-1638-002	PCA BACKPLANE IO GEN DISK	1	FR
040-2891-001	COVER INPUT POWER IO GEN	1	FE
040-2978-001	LATCH FIBRE OPTIC BULKHEAD IO	1	FE
040-3201-001	SHELF MODULE	1	FE
040-3240-001	COVER POWER CABLE I/O GEN	1	FE
040-3288-001	BRACKET CD ROM I-BRICK	1	FE
042-0708-003	GUIDE PIN DNET BR-NI .140DIA 2-FLAT	4	FE
042-0760-001	WASHER SHLDR INSUL .148ID .250OD	4	FE
042-0806-001	RACK EAR RIGHT DIE CAST	1	FE
042-0807-001	RACK EAR LEFT DIE CAST	1	FE
050-0616-001	HANDLE FAN PULL ¾U	3	FE
050-0620-001	PANEL TOP RIGHT	1	FE
050-0736-001	PANEL TOP LEFT I-BRICK	1	FE
050-0746-001	PANEL TOP CENTER I-BRICK	1	FE
050-0622-001	COVER CONTROL 3U	1	FE
050-0624-001	END CAP RIGHT	1	FE
050-0625-001	END CAP LEFT	1	FE
050-0683-001	LEVER SERVICE LEFT	1	FE
050-0684-001	LEVER SERVICE RIGHT	1	FE
050-0736-001	PANEL TOP LEFT	1	FE
050-0746-001	PANEL TOP CENTER	1	FE
060-0102-001	POWER SUPPLY D2D 48V INP 12V 10A	1	FR
060-0103-001	POWER SUPPLY D2D 48V INP 5V 25A	1	FR
060-0104-001	POWER SUPPLY D2D 48V INP 3.3V 30A	2	FR
060-0107-001	POWER SUPPLY D2D 48V INP 2.5V 30A	1	FR
064-0158-001	DRIVE CDROM 40X ATAPI	1	FE

Table 7-2 I-brick FRUs **(continued)**

Part Number	Description	Unit Qty	FE/FR[a]
9210201	BOARD PCI 66MHZ FC HSSDC COPPER	1	FR
9210218	BOARD GOLDENGATE 1394 TO IDE CONV	1	FE
9470407	DRIVE 18GB 10K RPM JBOD	1	FR
9470490	GASKET EMI FOAM .08X..16X17.25 UL94V	1	FE

[a]FE=field expense; indicates that the item may be scrapped. FR=field repairable; indicates that the component must be sent back for repair.

7.3 R Brick (013-3037-002)

Table 7-3 R-brick FRUs

Part Number	Description	Qty	FE/FR[a]
013-2801-001	FAN ASSY HOT SWAP R BRICK	2	FE
013-3040-002	ENCLOSURE ASSY ROUTER	1	FE
013-3041-002	COVER ASSY ROUTER	1	FE
013-3052-001	CABLE DOCK SHIELDED	8	FE
015-0353-001	HARNESS ASSY L1 INLET TO PWR BRD	1	FE
015-0355-001	HARNESS ASSY R INT TO L1/FANS	1	FE
015-0357-001	HARNESS ASSY R EXT L1 TO FANS/SC	1	FE
015-0368-001	HARNESS ASSY INTERNAL POWER ROUTER	1	FE
030-1439-002	PCA L1 SYSTEM CONTROL DISPLAY P1	1	FR
030-1631-002	PCA POWER ROUTER	1	FR
030-1632-001	PCA POWER INPUT 48V 30/2 A	1	FR
030-1634-002	PCA SN1 ROUTER	1	FR
040-3102-001	BRACKET MODULE MOUNTING RIGHT	1	FE
040-3103-001	BRACKET MODULE MOUNTING LEFT	1	FE
040-3174-001	COVER CONTROL R BRICK	1	FE
040-3201-001	SHELF MODULE	1	FE
042-0708-003	GUIDE PIN DNET BR-NI .140DIA 2-FLAT	16	FE
042-0760-001	WASHER SHLDR INSUL .148ID .250OD	16	FE
050-0626-002	COVER LEFT EAR ROUTER	1	FE

Table 7-3 R-brick FRUs (continued)

Part Number	Description	Qty	FE/FR[a]
050-0629-001	ROUTER CONTROL COVER	1	FE
060-0107-001	POWER SUPPLY D2D 48V INP 2.5V 30A	1	FR
9180127	CIRCUIT BREAKER 2.5A 50V UL CSA	1	FE

[a]FE=field expense; indicates that the item may be scrapped. FR=field repairable; indicates that the component must be sent back for repair.

7.4 P Brick (013-3034-002)

Table 7-4 P-brick FRUs

Pilot Part Number	Description	Qty	FE/FR[a]
013-2614-002	CARRIER ASSEMBLY PCI	12	FE
013-2800-001	FAN ASSY HOT SWAP 3-4U	3	FE
013-2882-001	CARRIER ASSY PWR P-BRICK	1	FE
013-2883-001	COVER TOP REAR ASSY P-BRICK	1	FE
013-2989-001	CARRIER P-BRICK PCI 12 SLOT	1	FE
013-3051-001	ENCLOSURE FRAME FRONT ¾U	1	FE
013-3052-001	CABLE DOCK SHIELDED	2	FE
013-3065-002	ENCLOSURE ASSY P BRICK	1	FE
015-0323-001	HARNESS ASSY L1 INLET TO PWR BRD	1	FE
015-0342-001	HARNESS ASSY PWR INLET TO PWR BOARD	1	FE
015-0356-001	HARNESS ASSY EXT TO FANS SC P1	1	FE
015-0358-001	HARNESS ASSY PIX INT L1 TO FANS	1	FE
030-1439-002	PCA L1 SYSTEM CONTROL DISPLAY P1	1	FR
030-1500-002	PCA POWER P-BRICK	1	FR
030-1569-003	PCA IO PCI 12 SLOT	1	FR
030-1583-001	PCA HOST INTERFACE P1	2	FR
030-1632-001	PCA POWER INLET 48V 30/2A	1	FR
040-2891-001	COVER INPUT POWER IO GEN	1	FE
040-3201-001	SHELF MODULE	1	FE
042-0708-003	GUIDE PIN DNET BR-NI .140DIA 2-FLAT	4	FE

Table 7-4 P-brick FRUs **(continued)**

Pilot Part Number	Description	Qty	FE/FR[a]
042-0760-001	WASHER SHLDR INSUL .148ID .250OD	4	FE
042-0806-001	RACK EAR RIGHT DIE CAST	1	FE
042-0807-001	RACK EAR LEFT DIE CAST	1	FE
050-0616-001	HANDLE FAN PULL	3	FE
050-0621-001	PANEL FRONT UPPER P-X BRICK	1	FE
050-0622-001	COVER CONTROL 3U	1	FE
050-0624-001	END CAP RIGHT	1	FE
050-0625-001	END CAP LEFT	1	FE
050-0683-001	LEVER SERVICE LEFT	1	FE
050-0684-001	LEVER SERVICE RIGHT	1	FE
060-0102-001	POWER SUPPLY D2D 48V INP 12V 10A	1	FR
060-0103-001	POWER SUPPLY D2D 48 INP 5V 25A	1	FR
060-0104-001	POWER SUPPLY D2D 48V INP 3.3V 30A	4	FR
060-0107-001	POWER SUPPLY D2D 48V INP 2.5V 30A	1	FR

[a]FE=field expense; indicates that the item may be scrapped. FR=field repairable; indicates that the component must be sent back for repair.

7.5 X Brick (013-3036-003)

Table 7-5 X-brick FRUs

Pilot Part Number	Description	Qty	FE/FR[a]
013-1637-003	I/O BLANKING BOARD ASSY	3	FE
013-2587-001	COVER ASSY XIO	1	FE
013-2800-001	FAN ASSY HOT SWAP 3-4U	3	FE
013-3051-001	ENCLOSURE FRAME FRONT ¾U	1	FE
013-3052-001	CABLE DOCK SHIELDED	2	FE
013-3056-001	ENCLOSURE ASSY XIO 4 SLOT	1	FE
013-3115-001	CARRIER ASSY PWR BRD XIO	1	FE
015-0323-001	HARNESS ASSY L1 INLET TO PWR BRD	1	FE
015-0342-001	HARNESS ASSY PWR INLET TO PWR BOARD	1	FE
015-0356-001	HARNESS ASSY EXT TO FANS SC P1	1	FE
015-0358-001	HARNESS ASSY PIX INT L1 TO FANS	1	FE
030-1439-002	PCA L1 SYSTEM CONTROL DISPLAY	1	FR
030-1565-002	PCA MIDPLANE XIO	1	FR
030-1566-002	PCA XTOWN INTERFACE XIO	1	FR
030-1571-002	PCA POWER I/X BRICK	1	FR
030-1632-001	PCA POWER INPUT 48V 30/2A	1	FR
040-2356-001	LOCKING TAB XIO	1	FE
040-3201-001	SHELF MODULE	1	FE
040-3210-001	COVER INPUT POWER XIO	1	FE
040-3240-001	COVER POWER CABLE IO GEN	1	FE
042-0708-003	GUIDE PIN DNET BR-NI .140DIA 2-FLAT	4	FE
042-0760-001	WASHER SHLDR INSUL .148ID .250OD	4	FE
042-0806-001	RACK EAR RIGHT DIE CAST	1	FE
042-0807-001	RACK EAR LEFT DIE CAST	1	FE
050-0616-001	HANDLE FAN PULL ¾U	3	FE
050-0621-001	PANEL FRONT UPPER P-X BRICK	1	FE
050-0622-001	COVER CONTROL 3U	1	FE
050-0624-001	END CAP RIGHT	1	FE
050-0625-001	END CAP LEFT	1	FE

Table 7-5 X-brick FRUs **(continued)**

Pilot Part Number	Description	Qty	FE/FR[a]
050-0683-001	LEVER SERVICE LEFT	1	FE
050-0684-001	LEVER SERVICE RIGHT	1	FE
060-0102-001	POWER SUPPLY D2D 48VRM 12V 10A	1	FR
060-0103-001	POWER SUPPLY D2D 48VRM 5V 25A	1	FR
060-0104-001	POWER SUPPLY D2D 48VRM 3.3V 30A	2	FR
060-0107-001	POWER SUPPLY D2D 48VRM 2.5V 30A	1	FR

[a]FE=field expense; indicates that the item may be scrapped. FR=field repairable; indicates that the component must be sent back for repair.

7.6 D Brick

Table 7-6 D-brick FRUs

Pilot Part Number	Description	Qty	FE/FR[a]
013-3219-001	DISK DRIVE ASSY 18GB 10K	VAR	FR
013-3220-002	DISK DRIVE ASSY 36GB 10K	VAR	FR
013-3221-001	DISK DRIVE ASSY 73GB 10K	VAR	FR
040-3206-001	SHELF ENCLOSURE DISK	1	FE
9470418	CABLE FC 1M (DB9 – DB9)	2	FE
9470391	FILLER DISK	VAR	FE
9470392	POWER SUPPLY 550 WATTS	2	FR
9470393	MODULE OPS	1	FR
9470394	MODULE I/O	2	FR
9470395	BLANKING MODULE (FILLER)	3	FE
9470396	CHASSIS 12 SLOT W/BACKPLANE	1	FR
9470404	KEY OPS LOCK	1	FE
9470406	KEY DRIVE LOCK	1	FE

[a]FE=field expense; indicates that the item may be scrapped. FR=field repairable; indicates that the component must be sent back for repair.

7.7 G Brick

Table 7-7 G-brick FRUs

Pilot Part Number	Description	Qty	FE/FR
013-1740-001	BLOWER ASSY KCAR	1	FR
013-1760-001	MIDPLANE ASSY WITH STIFFENERS	1	FE
013-2947-002	KCAR L1 CONTROLLER	1	FR
013-2954-001	PCA SET RM9 AND TM8-256	1	FR
013-3100-001	CABLE ASSY DIFF DNET 1M RED	VAR	FR
013-3229-001	PLENUM ASSY INTAKE ONYX 3	1	FE
018-0646-002	CABLE ASSY 13W3 TO 13W3 MALE 30FT	1	FE
018-0648-001	CABLE ASSY RADICAL BREAKOUT PCI	1	FE
018-0695-001	CABLE ASSY ADAT OPTICAL 2M	1	FE
018-0731-001	CABLE ASSY RADICAL PCI DIGITAL SYNC	1	FE
018-0942-001	CABLE ASSY USBA TO USBB 4POS 10FT	1	FE
018-0859-003	CABLE ASSY KCAR L1 DB9 USB L2	1	FE
018-0911-001	CABLE ASSY RJ45 TO RJ45 CAT5 30FT	1	FE
018-0925-001	CABLE ASSY BNC TO BNC RG59 30FT	1	FE
030-1439-002	PCA L1 SYSTEM CONTROL DISPLAY	1	FR
030-1593-001	PCA ASSY KTOWN2	1	FR
030-1649-001	PCA RADICAL AUDIO PCI	1	FR
030-1657-001	PCA PCI SIO	1	FR
040-3006-001	BRACKET FRONT LEFT ONYX3 RACK	1	FE
040-3007-001	BRACKET FRONT RIGHT ONYX3 RACK	1	FE
040-3008-001	BRACKET REAR LEFT ONYX3 RACK	1	FE
040-3009-001	BRACKET REAR RIGHT ONYX3 RACK	1	FE
040-3201-001	SHELF MODULE	1	FE
060-0026-002	POWER SUPPLY AC/DC 3100WDC	1	FR
062-0066-001	KEYBOARD USB USA W/BEEP	1	FE
063-0011-001	MOUSE 3 BUTTON USB	1	FE
9470545	HUB USB 4-PORT 100-240VAC	1	FE
9470546	HUB XTNDR USB-CAT5-USB 4PORT	1	FR

7.8 Cables

Table 7-8 Cable FRUs

Pilot Part Number	Description	Qty	FE/FR[a]
013-3100-001	CABLE ASSY DIFF DNET 1M RED	VAR	FR
013-3101-001	CABLE ASSY DIFF DNET 2M GRN	VAR	FR
013-3102-001	CABLE ASSY DIFF DNET 3M YEL	VAR	FR
013-3103-001	CABLE ASSY DIFF DNET 4M BLU	VAR	FR
018-0942-001	CABLE ASSY USBA TO USBB 4POS 10FT	VAR	FE
018-0784-001	CABLE ASSY FIBRECHANL HSSDC-DB9 3M	VAR	FE
018-0784-101	CABLE ASSY FIBRECHANL HSSDC-DB9 10M	VAR	FE
018-0928-001	CABLE ASSY 48V PWR 2M RS485	ONE	FE
042-0877-001	PLUG EMI D-NET CONNECTOR	VAR	FE

[a] FE=field expens; indicates that the item may be scrapped. FR=field repairable; indicates that the component must be sent back for repair.

7.9 Power Components

Table 7-9 Power Component FRUs

Pilot Part Number	Description	Qty	FE/FR
040-3382-001	POWER BAY MOUNTING BRACKET		
013-2895-001	PDU ASSY DUAL BAY 400V 3 PHASE IEC	1	FE
013-2896-001	PDU ASSY DUAL BAY 208/240V 3 PHASE	1	FE
013-2968-001	PDU ASSY 6 C14 OUTLET 250V 10 AMP	1	FE
013-3032-001	PDU ASSY SGL BAY 2 INPUTS HAR	1	FE
013-3074-001	PDU ASSY NEMA L6-30 250V 30A 1PH	1	FE
013-3135-001	POWER SUPPLY DPS 48V BAY	1OR/2	FR
018-0927-001	ABLE ASSY 48V PWR 1M RS485	VAR	FE
040-3201-001	SHELF MODULE	1	FE
040-3300-001	BRACKET MTG POWER BAY RIGHT REAR	1	FE
040-3301-001	BRACKET MTG POWER BAY LEFT REAR	1	FE
060-0119-001	POWER SUPPLY DPS 48V 1000W	VAR	FR

7.10 Misc. Support Equipment

Table 7-10 Support Equipment

Pilot Part Number	Description	QTY	FE/FR [a]
013-2568-003	SYSTEM CONTROLLER ASSY L2	1	FR
013-3106-001	ENCLOSURE ASSY L2 DISPLAY	1	FR
018-0905-001	CABLE ASSY L2 48V PWR ENTRY	VAR	FE
018-0782-001	CABLE ASSY L2 DISPLAY	1	FE
9350818	PWR CORD 250V 10A C14G JUMPER (ETHERNET)	1	FE
9470235	HUB ETHERNET 8PORT 10BASE-T NO BNC	1	FE
9470248	POWER SUPPLY 30W 220VAC 12VDC 2.5MM	1	FE
TBD	ROUTER, CISCO	1	FR
TBD	L3	1	FR

[a]FE=field expense; indicates that the item may be scrapped. FR=field repairable; indicates that the component must be sent back for repair.

7.11 Brick Hardware

Table 7-11 Brick Hardware

Part Number	Description
7260528	SCREW M4X14MM TORX PNH CRST CP STL
7260554	SCREW M4X8MM TRX PN SQR CP STL/ZIN
7260556	SCREW M3X6MM TRX PN SQR CP STL/ZIN
7260625	WASHER LOCK 1/4-40 INT TOOTH NI PLT
7260626	NUT 1/4-40 HEX NI PLT
7510114	SCREWLK 4-40 M/F 3/16HEX JKSCR QTY2

7.12 Rack and Trim, 39U

Table 7-12 Rack and Trim Components (39 U)

Part Number	Description	Qty
013-3164-002	POWER BAY COVER	
042-0835-001	DOOR LOCK	

Table 7-12 Rack and Trim Components (39 U) **(continued)**

Part Number	Description	Qty
013-3179-001	BRACKET ASSY GROUNDING RIGHT	
013-3180-001	BRACKET ASSY GROUNDING LEFT	
013-3199-001	SERVICE SHELF	
040-3018-001	BRACKET MOUNTING CONTROLLER LEFT	
040-3023-001	BRACKET MOUNTING CONTROLLER RIGHT	
040-3072-001	SHELF UTILITY	
040-3112-001	DOOR ASSY FRONT	
040-3113-001	DOOR ASSY REAR	
040-3114-001	PANEL SIDE REAR	
040-3115-001	BRACKET HINGE FRONT	
040-3116-001	BRACKET LATCH FRONT	
040-3117-001	BRACKET HINGE REAR	
040-3118-001	BRACKET LATCH REAR	
040-3119-001	PANEL TOP FRONT	1
040-3221-001	RAIL MOUNTING SHELF RIGHT	2
040-3222-001	RAIL MOUNTING SHELF LEFT	2
050-0581-001	GRILL DOOR BLUE	1
050-0581-101	GRILL DOOR BLACK	1
050-0723-001	COMB CABLE SN1 RACK	
050-0724-001	DIVIDER CABLE COMB SN1 RACK	
050-0734-001	PANEL LEFT SIDE	
050-0735-001	PANEL RIGHT SIDE	
050-0745-001	FILLER PANEL 1U	
050-0803-001	PANEL TRIM 3U FILLER	
7260449	TINI-JAX MINIATURE PHONE JACK	2
7260472	CASTER SWIVEL 2.5 DIA 400LB	2
7260597	LEVELER ADJ M12 X 102MM STL/ZN	4
7260667	WHEEL 6.00DIA X 2.00W 1250LB POLY	2
7260674	WASHER FLAT M8 STL/ZN	8
7260675	WASHER LK M8 HELICAL SPR STL/ZN	8

Table 7-12 Rack and Trim Components (39 U) **(continued)**

Part Number	Description	Qty
7260676	NUT HEX M8 STL/ZIN STYLE1	8
7260677	NUT HEX M12 STL/ZN STYLE1	4
7260692	WASHER LK M12 HELICAL SPR STL/ZIN	2
7260693	SCREW M12X90MM CAP HEX SS	2
7260766	FASTENER PUSH IN 7/16OD .30DIA NYL	12
7260841	SCREW M5X30MM TRX PN SQR CP STL/ZN	56
7260846	WASHER FLT .209IDX.438ODX.06 STL/ZN	56
7260847	SPACER .201IDX5/16ODX.687L STL/ZN	56

7.13 Rack and Trim, 17U

Table 7-13 Rack and Trim Components (17 U)

Part Number	Description	Qty
040-3185-001	RAIL MOUNTING SHELF SHORT RIGHT	2
040-3186-001	RAIL MOUNTING SHELF SHORT LEFT	2
040-3181-001	PANEL RACK SHORT TOP	1
040-3182-001	PANEL RACK SHORT RIGHT	1
040-3183-001	PANEL RACK SHORT LEFT	1
7260542	CASTER SWIVEL 2.5 DIA 140 LBS	2
7260543	CASTER RIGID 2.5 DIA 140 LBS	2
7260597	LEVELER ADJ M12 X 102MM STL/ZN	4
7260677	NUT HEX M12 STL/ZN STYLE1	4
7260683	WASHER FLAT M6 STL/ZN	16
7260684	WASHER LK M6 HELICAL SPRING STL/ZN	16
7260685	NUT HEX M6 STL/ZN STYLE1	16
7260698	SCREW M5X10MM TRX PN SQR CP STL/ZN	24
7260766	FASTENER PUSH IN 7/16OD .30DIA NYL	12

7.14 PCI Options

Table 7-14 PCI Options

Part Number	Mrktg Code	Description	FE/FR[a]
030-0950-002	PCI-AUD-8C-ALL	PCA RADICAL AUDIO PCI	FR
9210109	PCI-FDDI-DAS	BOARD PCI FDDI DAS	FR
9210129	PCI-GIGENET-HW	BOARD PCI GIGABIT ETHERNET ADAPTER	FR
9210138	PCI-ATM-OC12	BOARD PCI ATM OC-12 ADAPTER BOARD	FR
9210141	PCI-ATMOC3-1P	BOARD PCI ATM OC-3SC ADAPTER	FR
9210185	PCI-SCSI-HVDF-2P	BOARD SCSI CONT ULTRA DIFFERENTIAL	FR
9210186	PCI-SCSI-SE-2P	BOARD SCSI CONT ULTRA SGL ENDED	FR
9210187	PCI-SCSI-LVDF-2P	BOARD SCSI CONT ULTRA2 SGL/DIFF	FR
9210200	PCI-FC-1PCOP-A	BOARD PCI 66MHZ FC OPTICAL	FR
9210201	PCI-FC-1POPT-A	BOARD PCI 66MHZ FC HSSDC COPPER	FR

[a]FE=field expense; indicates that the item may be scrapped. FR=field repairable; indicates that the component must be sent back for repair.

7.15 XIO Options

Table 7-15 XIO Options

Part Number	Mrktg Code	Description	FE/FR
013-2476-001	XT-HD-HW	ASSY XT-HD	FR
030-0968-004	XT-HIPPI-800S	PCA HIPPI SERIAL O2K	FR
030-1213-004	560-XT-VME-9U	PCA CROSSTOWN TO VME I/F VME 9U	FR
030-1221-004	560-XT-VME-6U	PCA CROSSTOWN TO VME I/F VME 6U	FR
030-1305-002	XT-DIVO	PCA DIVO I/O DVC-READY	FR
030-1361-003	XT-GSN-C-1XIO	PCA XT-GSNIC SUPER-HIPPI BOARD	FR

7.16 Tools List

Table 7-16 Tools List

Part Number	Description	Cost $
018-0971-001	L1 SERVICE CABLE	
018-0855-001	CABLE ASSY LOOPBACK P1 DNET SN1	425
7260422	TORX DRIVER BALL-POINT T-20	5.09
7260423	TORX DRIVER BALL-POINT T-25	5.23
7260650	TORX DRIVER BALL-POINT T-10	4.96
7260653	NUTDRIVER, 3/16" HEX HOLLOW SHAFT	3.43
7260654	TORX DRIVER T-8	3.42
7260655	EXTENSION, 6 INCH 3/8" DRIVE	6.45
7260726	SOCKET 13MM STANDARD 3/8" DRIVE	3.39
7260744	WRENCH QUICK 13MM	8.12
7260775	RATCHET, REVERSIBLE 3/8 INCH DRIVE	23.04
7260788	HOLDER CURVED TIP	8.68
7260816	WRENCH 5/16 INCH COMBINATION	10.00
7450122	PLUG FEM 3P 4W 60A 250VAC (JAPAN ONLY)	268.52
8050126	STRAP WRIST ADJ WITH 10FT CORD	30.08
8050127	SERVICE KIT ESD PORTABLE	46.01
9470540	SCREWDRIVER SLTD 7/32 TIP 3IN BLADE	2.10
9470544	SCREWDRIVER SLTD ¼ TIP 6IN BLADE	3.19
9470556	LEVEL 9" MAGNETIC TORPEDO	7.62
9470557	CASE TOOL COMPACT ZIP	33.00
9470575	SOCKET 17MM STANDARD 3/8: DRIVE	3.65
9470610	CUTTER DIAGONAL 6"	10.50
9470611	RCPT AC 6-15R 15A 250V (JAPAN ONLY)	22.80
9470618	SOCKET 19MM STANDARD 3/8" DRIVE	3.78
9470619	SCREWDRIVER PHILLIPS #1	2.48
———————	SMOCKS – CHOOSE TO FIT	————
8050110	SMOCK SMALL TEAL W/SGI LOGO	23.24
8050111	SMOCK MEDIUM TEAL W/SGI LOGO	23.24
8050112	SMOCK LARGE TEAL W/SGI LOGO	23.24
8050113	SMOCK LARGE TEAL W/SGI LOGO (TALL)	25.26

Table 7-16 Tools List **(continued)**

Part Number	Description	Cost $
8050114	SMOCK EXTRA LARGE TEAL W/SGI LOGO	23.24
8050115	SMOCK EXTRA LARGE TEAL W/SGI LOGO (TALL)	25.26
8050116	SMOCK XX-LARGE TEAL W/SGI LOGO	25.26
8050117	SMOCK XX-LARGE TEAL W/SGI LOGO (TALL)	26.98
8050118	SMOCK XXX-LARGE TEAL W/SGI LOGO	29.00
8050119	SMOCK XXX-LARGE TEAL W/SGI LOGO (TALL)	29.00
8050120	SMOCK XXXX-LARGE TEAL W/SGI LOGO	29.31
8050121	SMOCK XXXX-LARGE TEAL W/SGI LOGO (TALL)	30.31

Notes